

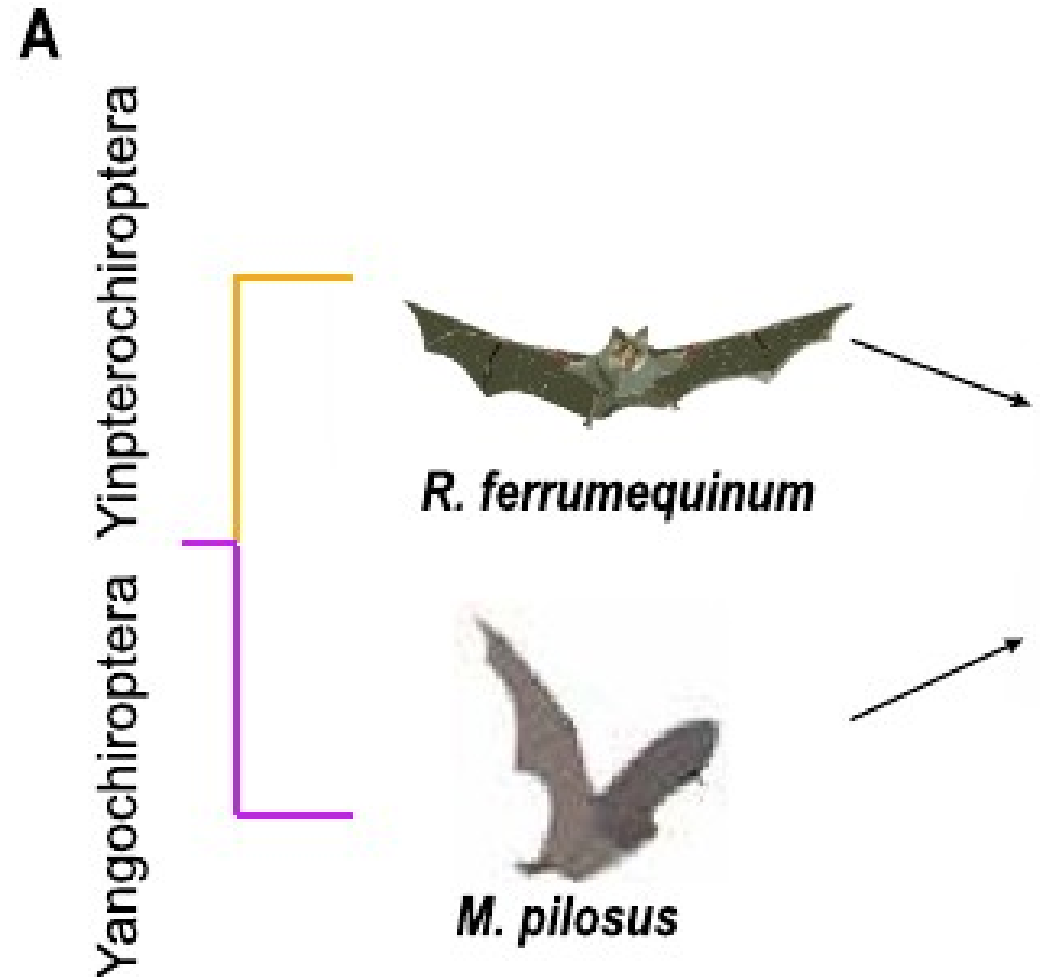
Lecture 16

Bat echolocation I: CF-FM bats and Doppler- shift compensation (Suga)

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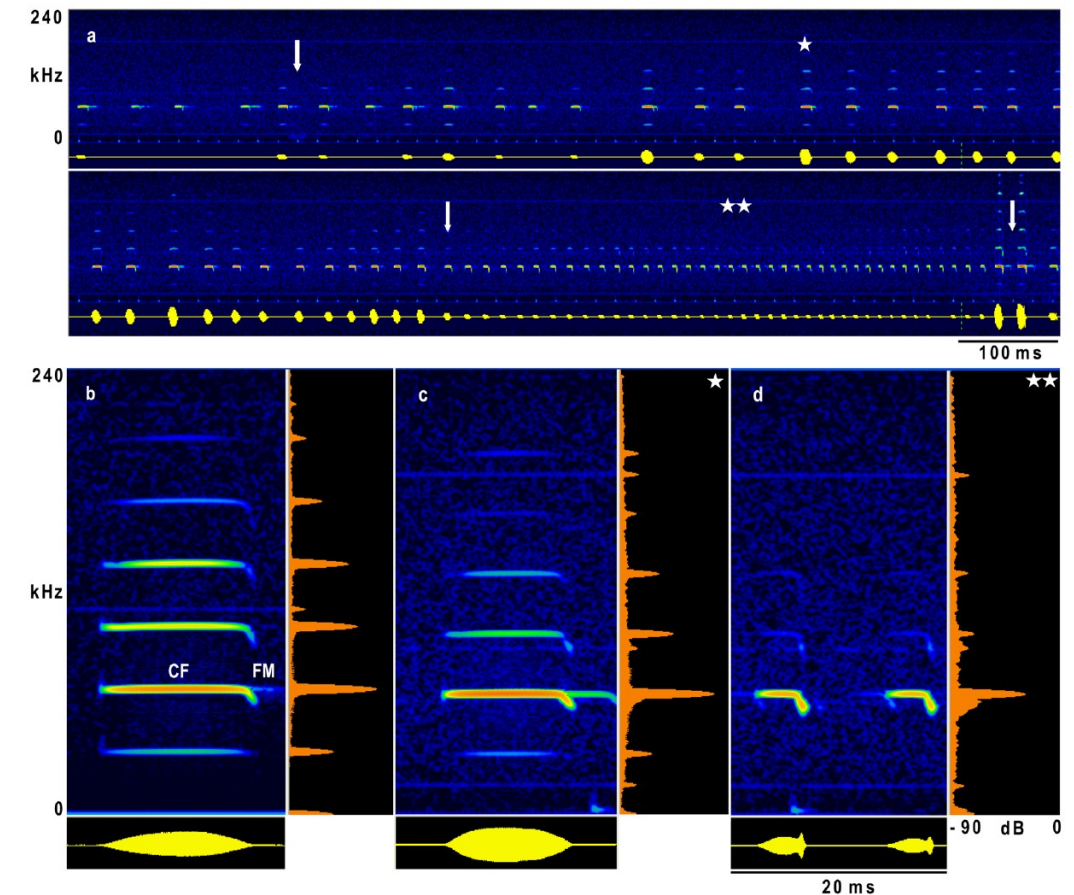
Wang et al. (2025), Fig. 1(A). CF and FM study bats.

CF-FM calls combine a sustained tone and a short sweep

Echolocation is active sensing with emitted sounds and returning echoes. A **constant-frequency component**, or CF, maintains nearly one frequency; a **frequency-modulated component**, or FM, changes frequency during the signal. CF-FM bats combine a prolonged tone with a brief downward sweep.

In *Hipposideros armiger*, recordings during flight contain several **harmonics**, frequency components at integer multiples of the lowest component. The second harmonic carries the greatest amplitude. Its CF frequency is around 65–66 kHz in the animals studied by Schoeppler and colleagues.

The CF component persists during approach to a landing target, even as call duration and harmonic prominence change. A narrowband carrier and a changing terminal sweep therefore coexist within individual calls; the signal is not simply a pure tone.



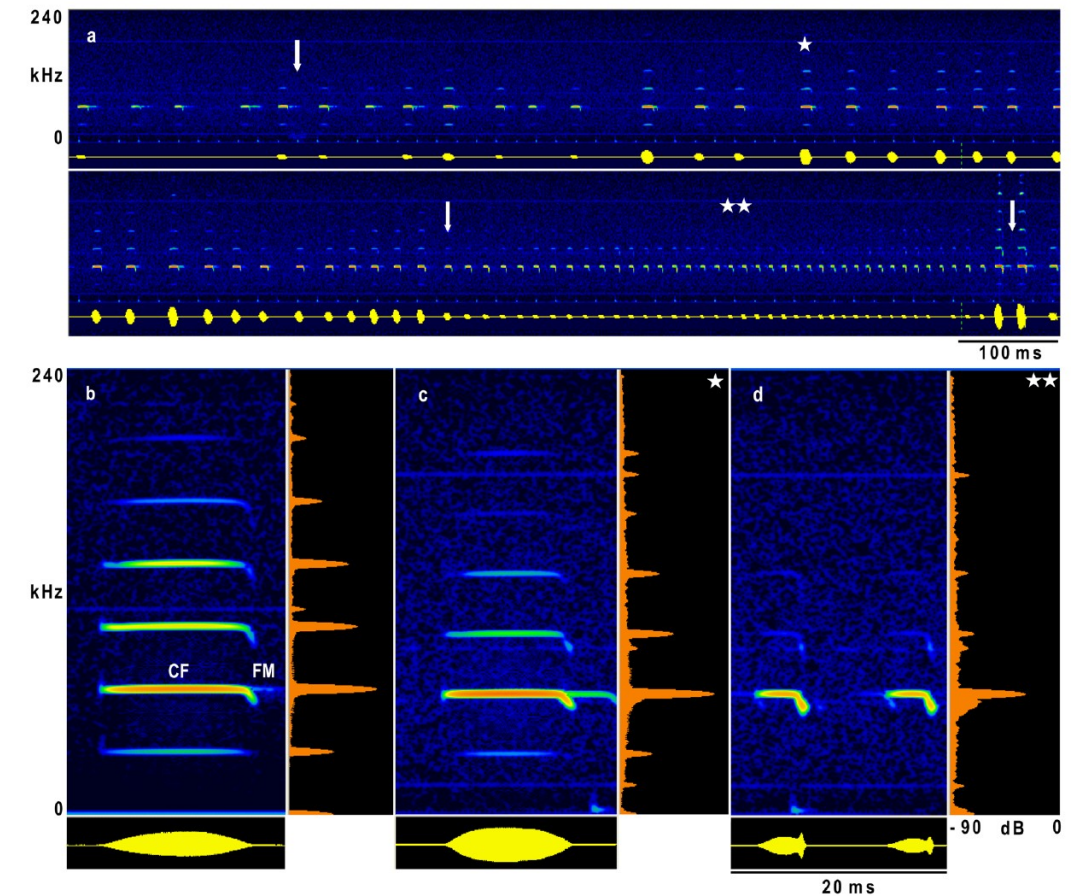
Schoeppler et al. (2018), Fig. 1(a–d). Flight call sequence and CF-FM structure, in kHz and ms.

Flutter changes the echoes carried by the CF component

Flutter is the periodic motion of an insect's wings. Flutter-detecting bats receive echoes whose amplitude and frequency vary as the reflecting wings move. The long CF component supplies a sustained carrier on which those changes can occur.

The **auditory fovea** is an expanded, sharply tuned representation of a narrow frequency range in the hearing system. In CF-FM bats, that range corresponds closely to the returning CF component rather than to all ultrasonic frequencies equally.

Flight-induced shifts can move the echo carrier away from this specialized range. Doppler-shift compensation keeps echoes from stationary targets ahead near the preferred frequency, supporting analysis of smaller variations associated with flutter. The behavioral adjustment and peripheral specialization are complementary adaptations.



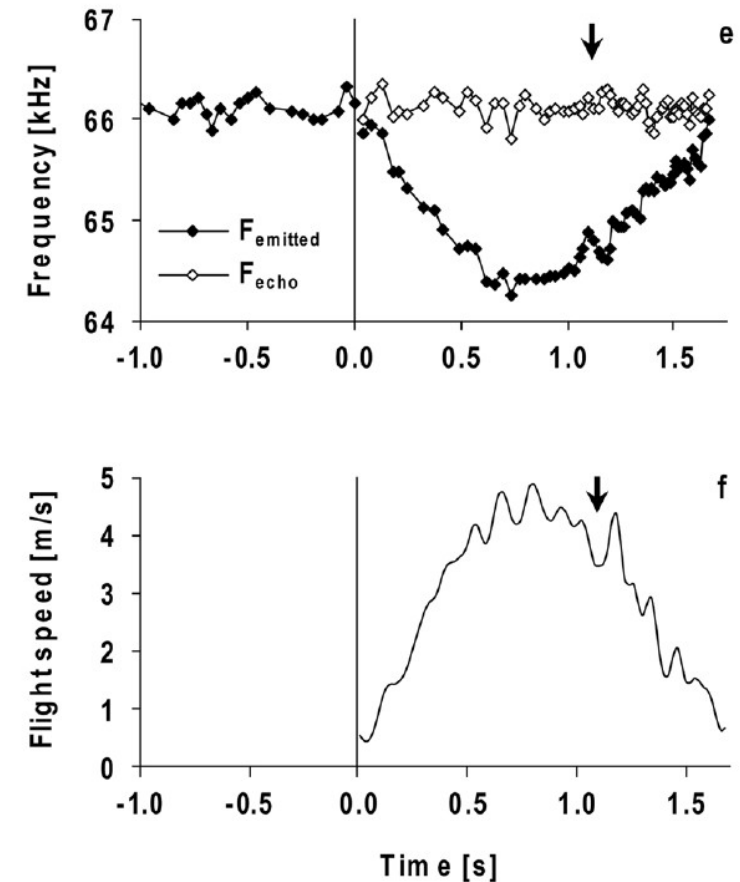
Schoeppler et al. (2018), Fig. 1(a–d). Flight call sequence and CF-FM structure, in kHz and ms.

Approaching a reflector raises returning echo frequency

A **Doppler shift** is a frequency change caused by relative motion between a sound source, a reflector, and a receiver. When a bat approaches a stationary object, the returning echo has a higher frequency than the emitted sound.

Doppler-shift compensation lowers the frequency of subsequent calls as approaching motion raises echo frequency. The regulated quantity is the returning CF frequency. The emitted CF frequency must therefore change rather than remain fixed throughout flight.

Schoeppler and colleagues recorded calls and reconstructed flight toward a grid. Call frequency fell during acceleration and recovered during deceleration. Schuller and colleagues independently manipulated echo frequency in stationary bats, separating an auditory-feedback effect from the mechanical effects of flight.



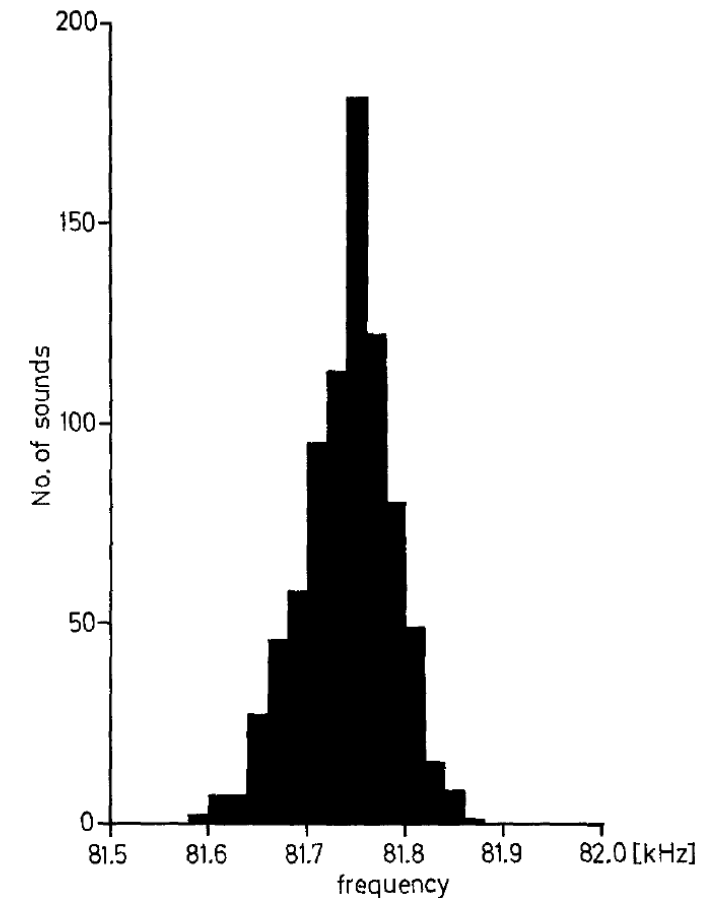
Schoeppler et al. (2018), Fig. 2(e–f). Measured emission and flight speed, with calculated echo frequency.

Resting frequency is an individual calling baseline

The **resting frequency** is the mean CF frequency of consecutive calls emitted by a stationary bat. In Schuller and colleagues' greater horseshoe bats, *Rhinolophus ferrumequinum*, individual resting frequencies ranged from about 81 to 84.2 kHz.

The investigators measured the central CF portion, excluding the first and last 5 ms to avoid the adjoining FM sweeps. Repeated calls clustered closely around an individual's baseline. Playing back unshifted calls did not systematically change that baseline.

Resting frequency is therefore a reference for describing changes in emitted frequency, not the frequency at which all compensated echoes must return. Individual and temporal differences require a baseline measured in the same animal and experimental period.



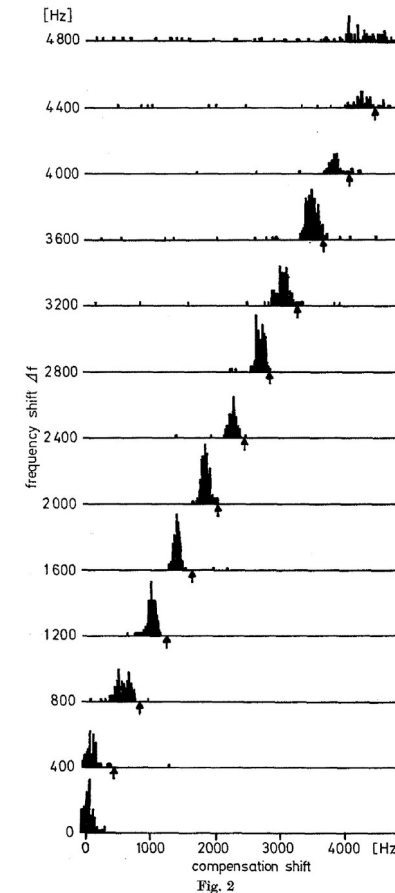
Schuller et al. (1974), Fig. 1. Distribution of CF frequencies in a stationary horseshoe bat.

Shifted playback causes subsequent calls to fall in pitch

Schuller and colleagues returned each horseshoe bat's own call through an electronic frequency shifter. The stationary animal could move its head, but it did not fly. The microphone and loudspeaker were positioned 34 cm in front of it.

When the artificial echo was shifted upward, later emitted calls decreased in CF frequency. After several calls, the animal reached a new emission level. Increasing the imposed positive shift required progressively greater lowering of the outgoing frequency.

This **closed-loop playback** preserves the link between the animal's current call and its feedback. The compensation response depends on an experimentally imposed frequency discrepancy, rather than requiring actual motion toward a reflector.



Schuller et al. (1974), Fig. 2. Emission-frequency compensation under increasing positive echo shifts, in Hz.

Compensated echoes sit slightly above resting frequency

The **reference frequency** is the returning CF frequency maintained during compensation. In Schuller and colleagues' experiments, the downward change in emission was slightly smaller than the upward shift imposed on the artificial echo.

The resulting echo therefore remained above resting frequency. This **compensation offset**, the difference between reference and resting frequencies, was approximately 50–300 Hz across the bats studied and stayed relatively constant across an individual's compensation range.

Compensation does not mean returning every echo to the stationary emission baseline. It means controlling feedback near an individual reference frequency. A small persistent offset and a large adjustable change in emitted frequency can occur together.

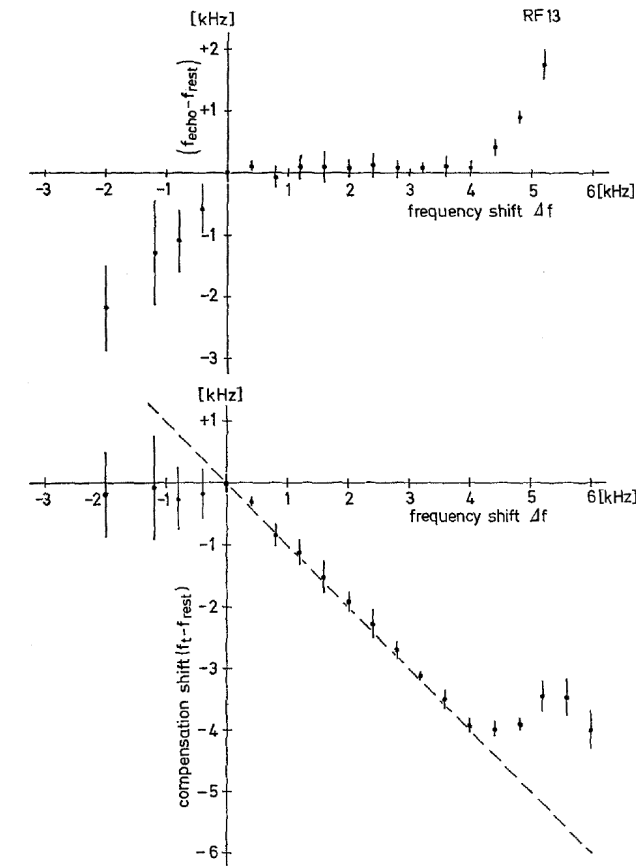


Fig. 3

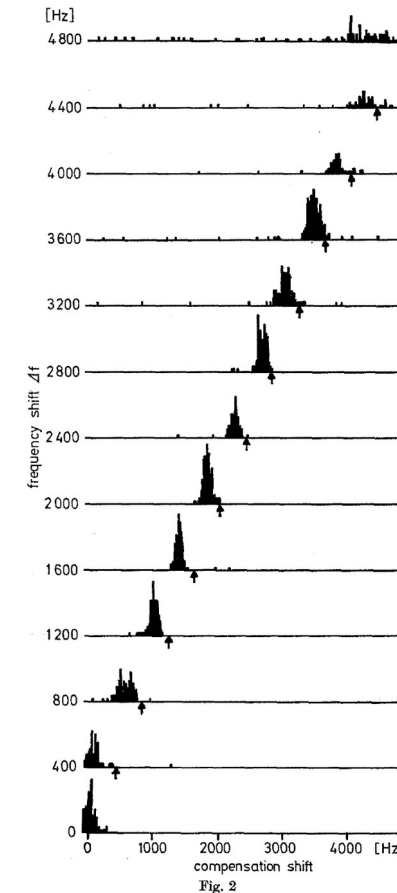
Schuller et al. (1974), Fig. 3. Echo-frequency offset and emission compensation across imposed shifts, in kHz.

Early playback tests found little response to lower echoes

In the 1974 experiments, downward shifts of artificial echoes did not evoke a systematic increase in emitted frequency. The bats continued calling near resting frequency, although their calls became somewhat less regular in frequency.

This finding supported an initially asymmetric account: echoes above the reference band lowered emission, whereas echoes below it appeared ineffective. However, the intensity dependence of the response was not systematically measured in that study.

Metzner and colleagues later tested feedback above and below resting frequency at comparable levels above hearing threshold. Their results changed the interpretation of the earlier asymmetry: a weak response to lower-frequency feedback does not establish that this feedback is absent from vocal control.



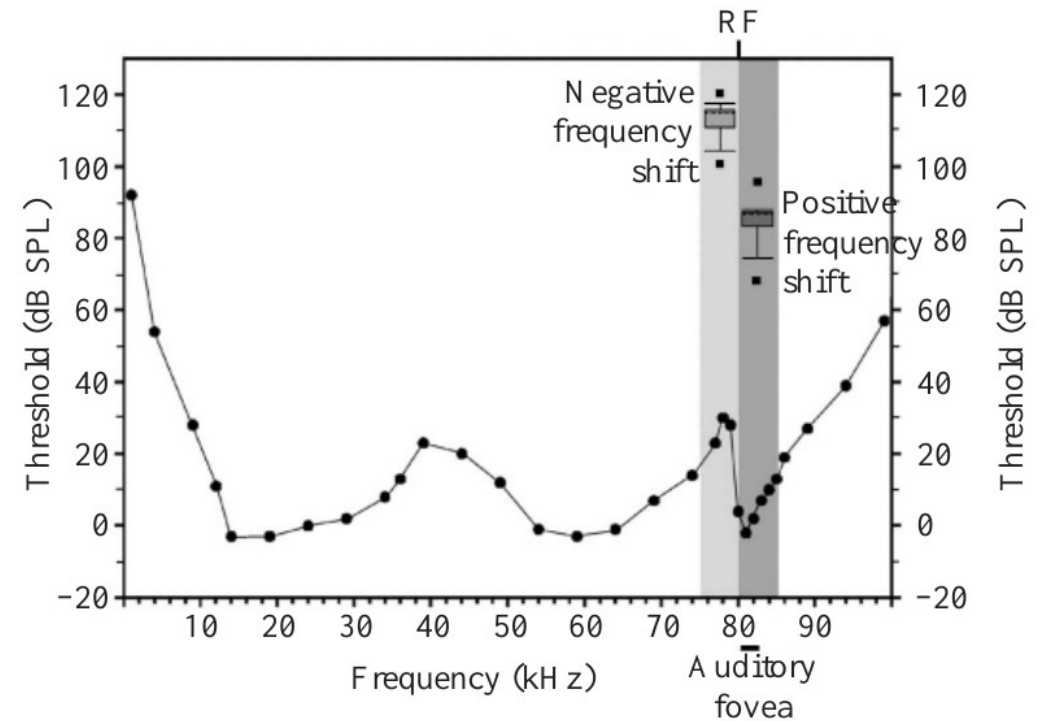
Schuller et al. (1974), Fig. 2. Emission-frequency compensation under increasing positive echo shifts, in Hz.

Equal sound levels need not produce equal audibility

An **audiogram** relates sound frequency to the minimum level required for detection. The horseshoe bat's auditory fovea has a low threshold near its preferred echo band, with substantially poorer sensitivity at some lower frequencies.

Sound pressure level, or dB SPL, describes physical sound amplitude. **Sensation level** describes how far a sound lies above the hearing threshold at its frequency. Equal dB SPL values can correspond to different sensation levels.

Metzner and colleagues delivered positive and negative frequency shifts at approximately comparable suprathreshold levels. Lower-frequency feedback required louder playback. This control separated the sign of a frequency shift from the reduction in audibility that otherwise accompanies a shift away from the fovea.



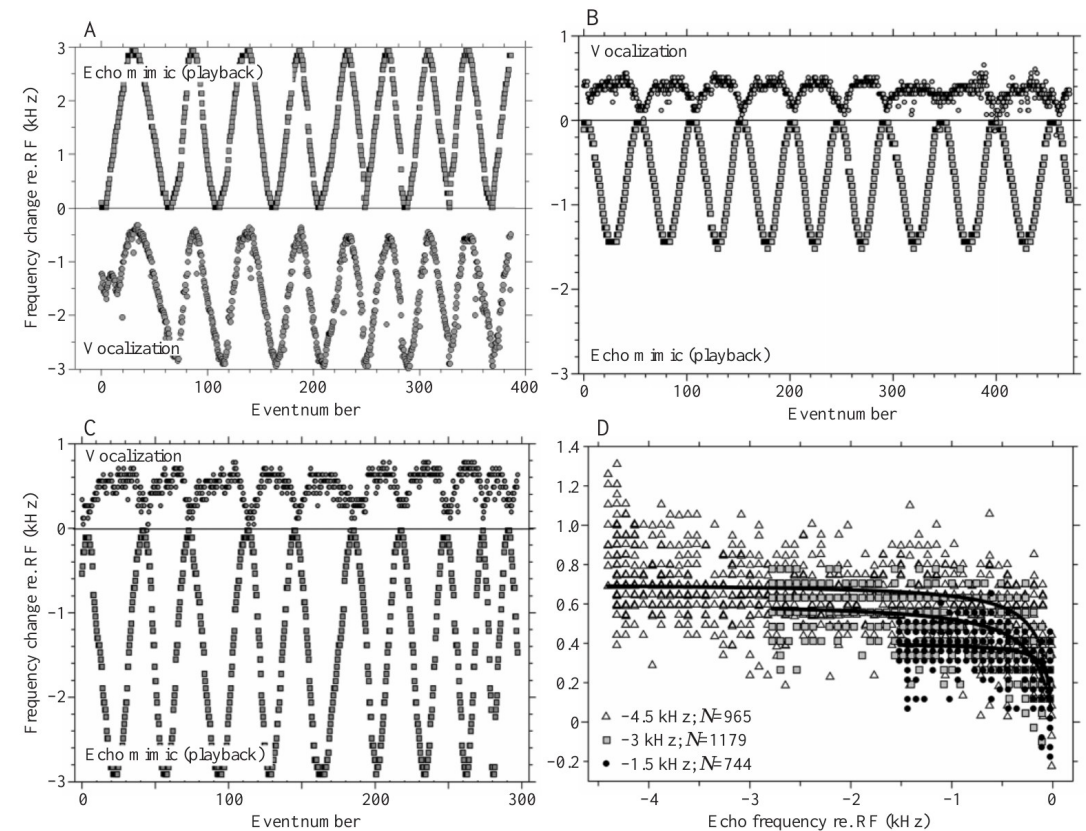
Metzner et al. (2002), Fig. 1. Auditory thresholds and playback ranges; audiogram credited in the paper to Long and Schnitzler (1975).

Audible lower-frequency echoes raise emitted frequency

Metzner and colleagues sinusoidally varied the frequency of feedback made from each bat's own call. A **sinusoidal modulation** is a repeated smooth rise and fall. Feedback shifts extended above or below the animal's resting frequency.

Positive shifts caused emitted frequency to decrease. At sufficient playback levels, negative shifts caused emitted frequency to increase above resting frequency. The opposite changes followed the changing frequency of the artificial echo.

The vocal system therefore receives useful feedback from both sides of the resting band. This two-direction response contradicts an explanation in which lower-frequency echoes have no influence on call frequency. It does not imply equally complete correction in the two directions.



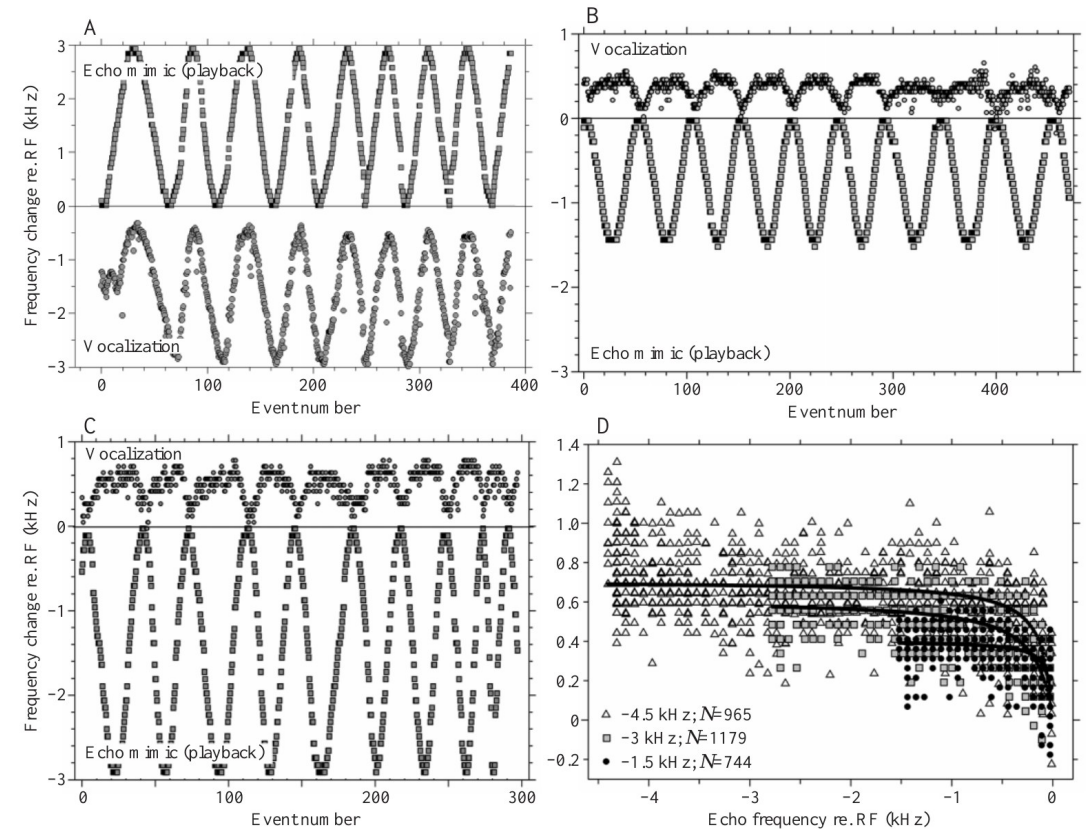
Metzner et al. (2002), Fig. 2(A-D). Call and playback changes for positive and negative frequency shifts.

Upward vocal corrections are smaller than downward ones

In Metzner and colleagues' playback experiments, bats compensated for roughly 95% of positive shifts but only up to about 22% of the range of negative shifts. Increasing emission above resting frequency was possible but comparatively limited.

A negative shift as large as 4.5 kHz produced a maximum observed emission increase of 1.51 kHz. Detecting feedback and generating a full motor correction therefore cannot be treated as the same capacity.

The authors related the asymmetry to restrictions on vocal production and discussed constraints involving the larynx. Their behavioral data establish unequal correction in the two directions. They do not independently isolate a particular muscle or biomechanical limit as its sole cause.



Metzner et al. (2002), Fig. 2(A–D). Call and playback changes for positive and negative frequency shifts.

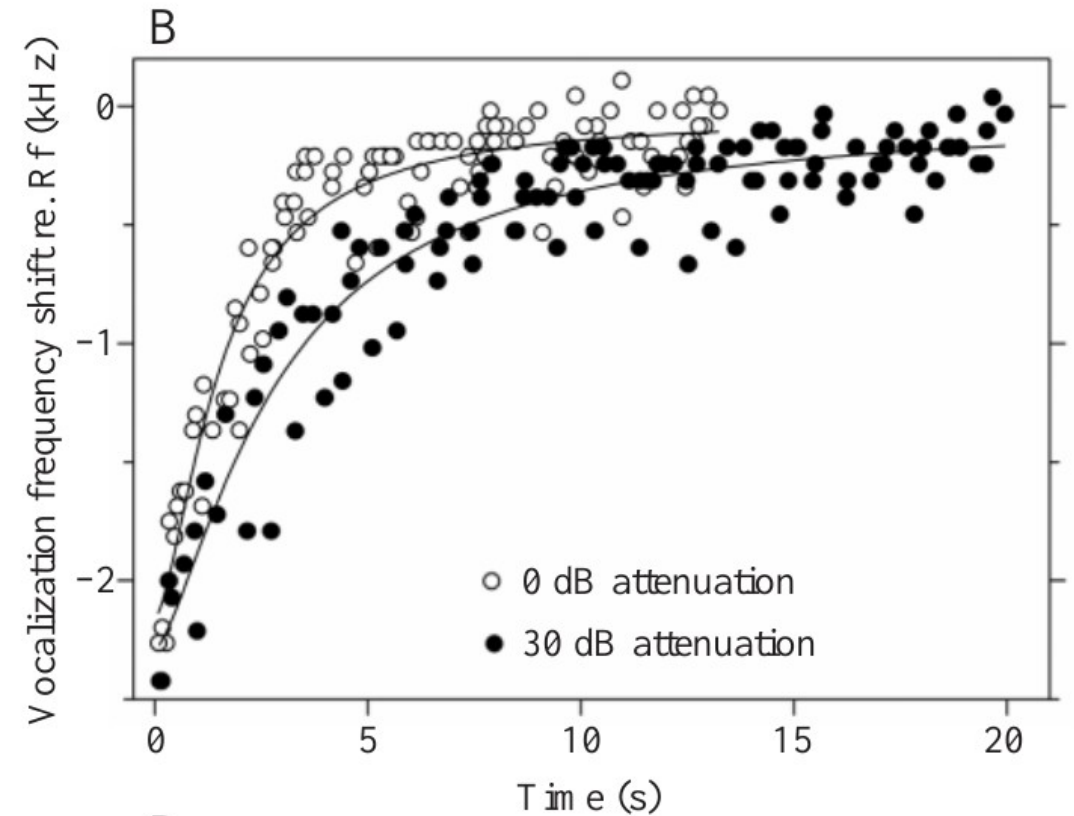
Louder feedback accelerates correction in both directions

Metzner and colleagues varied echo-mimic attenuation while maintaining stepwise frequency perturbations.

Attenuation is a reduction in signal level. Stronger feedback produced faster vocal adjustment for both upward and downward corrections.

For recovery after a negative step, reported response time constants shortened from about 2.28 s at 30 dB attenuation to 0.89 s at zero attenuation. For positive steps, corresponding values shortened from about 1.64 s to 0.75 s.

The **time constant** describes the measured progress toward the adjusted call frequency. These differences require the control response to depend on feedback strength as well as frequency. A sign-only rule, insensitive to audibility, cannot account for the observed response speeds.



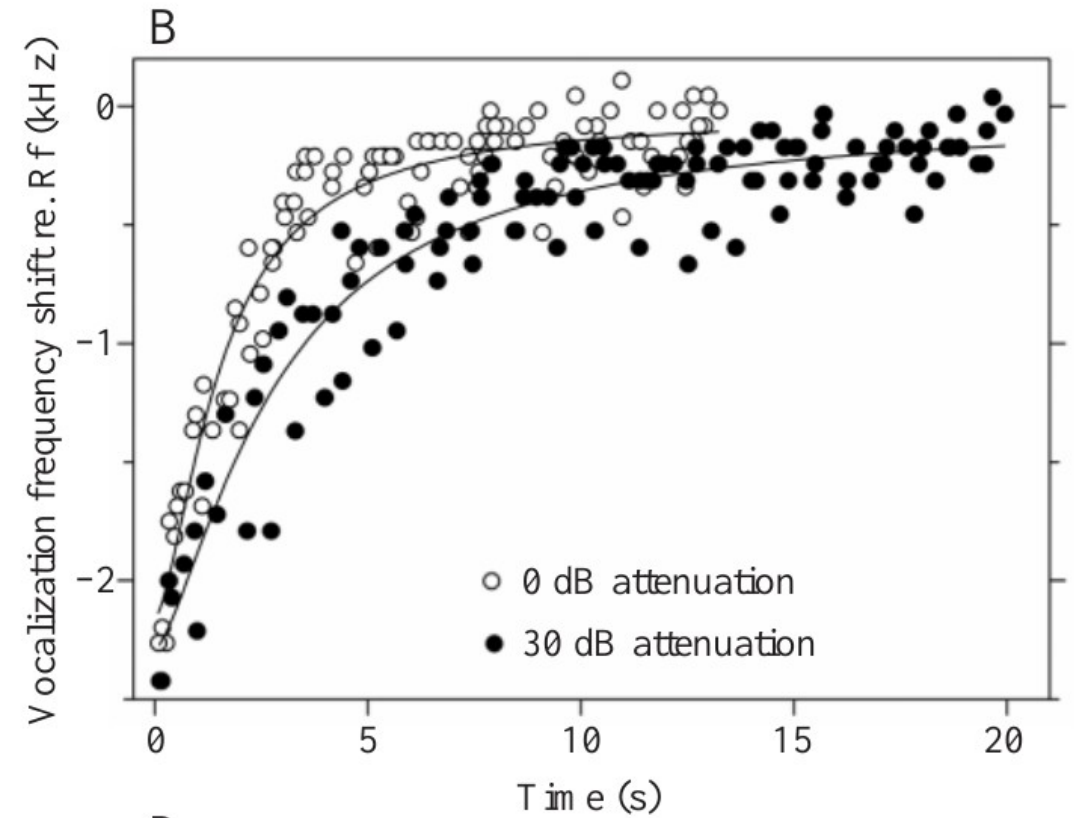
Metzner et al. (2002), Fig. 3(B). Call-frequency recovery under strong and attenuated feedback, in seconds.

Vocal correction is measured at the acoustic output

An auditory–vocal feedback experiment measures a sensory intervention and a change in emitted sound. Metzner and colleagues shifted echo frequency and varied playback intensity in stationary horseshoe bats. The acoustic output changed even without flight, establishing a feedback contribution to call-frequency control.

At comparable levels above hearing threshold, higher-frequency feedback lowered subsequent emissions and lower-frequency feedback raised them. Correction became faster as echoes became more audible. These measurements link an auditory discrepancy to a vocal adjustment but leave the intervening neural and muscular steps unresolved.

The experiment did not image the vocal apparatus, record its motor neurons or isolate a muscle-specific mechanism. Its established endpoint is a change in CF emission, measured in frequency and over time. A precise acoustic correction should therefore not be mistaken for a complete anatomical account of the feedback circuit.



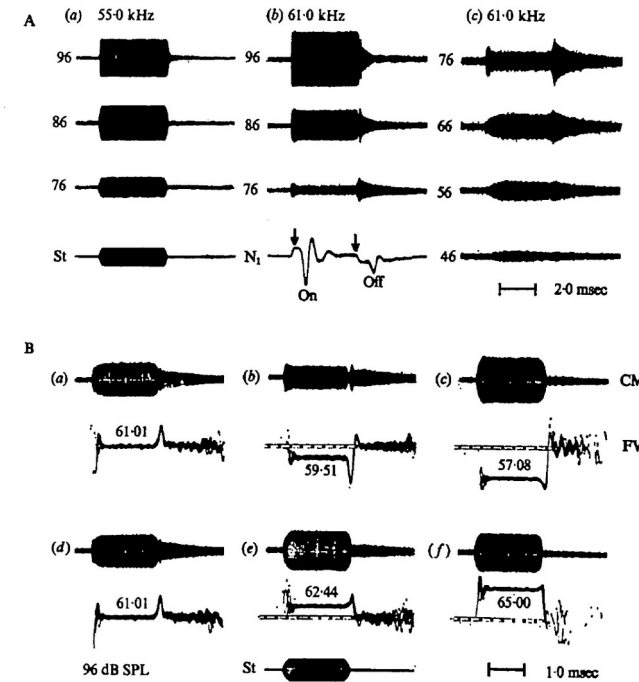
Metzner et al. (2002), Fig. 3(B). Call-frequency recovery under strong and attenuated feedback, in seconds.

Cochlear recordings separate receptor and neural signals

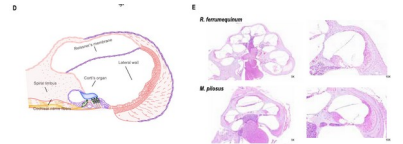
The **cochlea** is the inner-ear structure that converts sound into receptor activity. Suga and colleagues recorded **cochlear microphonics**, or CM, electrical responses related to acoustic stimulation of the cochlea, using an electrode at the round window.

They also recorded the **N1 response**, summed activity of primary auditory neurons. CM and N1 therefore sampled different stages of peripheral processing: receptor-related activity and neural discharge. Tone bursts allowed the researchers to compare responses during sound and at its onset and cessation.

The paper's short CM stimuli lasted 4 ms, with rapid rise and decay. Stimulus frequency and sound pressure were controlled at the ear. These recordings test whether narrow frequency selectivity and offset responses already arise peripherally, before a cortical interpretation of echo information.



Suga et al. (1975), Fig. 1(A–B). Round-window cochlear and neural responses to calibrated tone bursts.



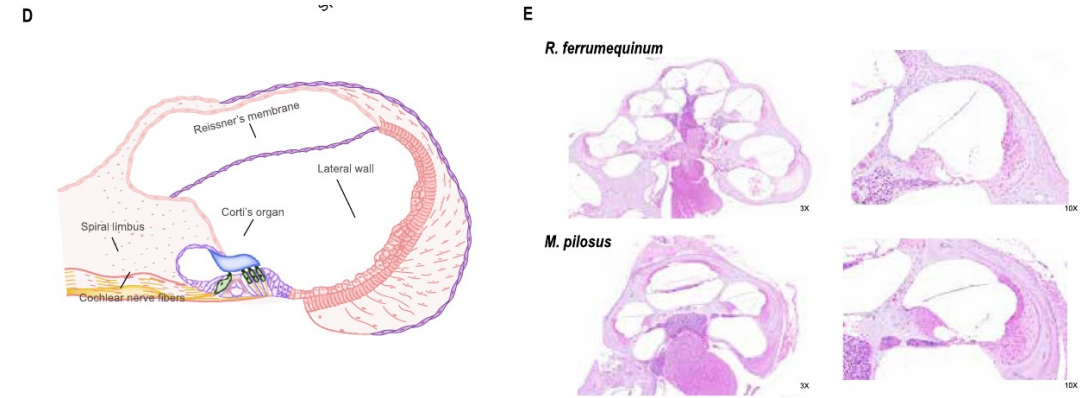
Wang et al. (2025), Fig. 1(D–E).
Published cochlear anatomy
and stained tissue sections.

Cochlear anatomy links vibration to neural output

The **cochlea** is the coiled inner-ear structure that converts acoustic vibration into sensory activity. Wang and colleagues combined single-nucleus sequencing with hematoxylin-and-eosin tissue staining in the CF bat *Rhinolophus ferrumequinum* and the FM bat *Myotis pilosus*. Their published sections locate sensory tissue, neural tissue and the cochlear lateral wall.

The **neurosensory epithelium** contains hair cells and supporting cells. Hair cells provide the sensory interface; spiral ganglion neurons carry cochlear information toward the brainstem. The atlas resolved 14 cell types in the horseshoe bat and 13 in the FM bat, distinguishing several cooperating tissue compartments rather than treating the cochlea as one uniform population.

Anatomical localization and molecular classification describe where candidate mechanisms reside. Neither tissue staining nor RNA sequencing measured Doppler compensation during flight. Their contribution is a cellular framework for interpreting the peripheral specialization established by acoustic and electrophysiological experiments.



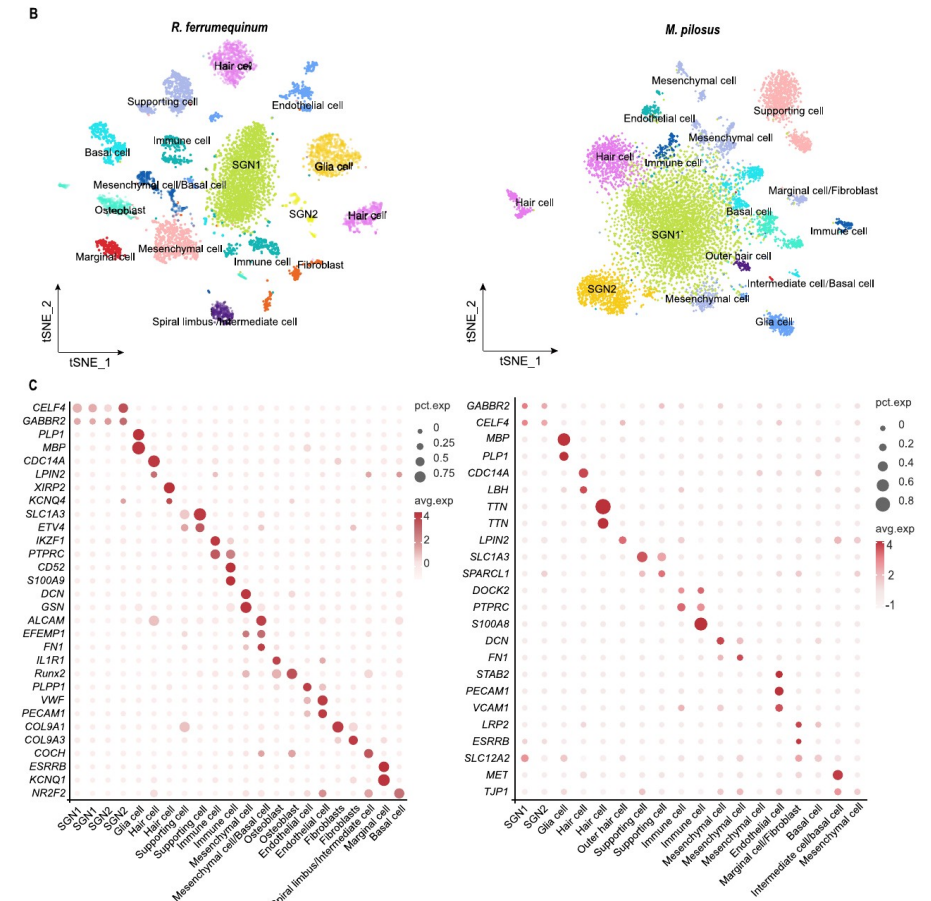
Wang et al. (2025), Fig. 1(D–E). Published cochlear anatomy and stained tissue sections.

Nuclear RNA identifies distinct cochlear cell populations

Single-nucleus RNA sequencing measures RNA associated with individual cell nuclei. A **marker gene** helps identify a cell population by its expression pattern. Wang and colleagues grouped cochlear nuclei using expression profiles and used more than 200 marker genes to support the identification of neural, sensory, immune and structural populations.

Their species-specific analyses resolved 14 cell types in *Rhinolophus ferrumequinum* and 13 in *Myotis pilosus*. Neural populations included two spiral-ganglion categories and glial cells. Hair cells and supporting cells occupied the sensory compartment, while marginal and basal cells contributed to lateral-wall categories.

The plotted expression clusters are molecular groupings, not a physical map of cochlear position or a measured frequency-tuning map. A cell's RNA profile can identify candidate functions, but does not establish its firing threshold, preferred echo frequency or causal contribution to vocal feedback.



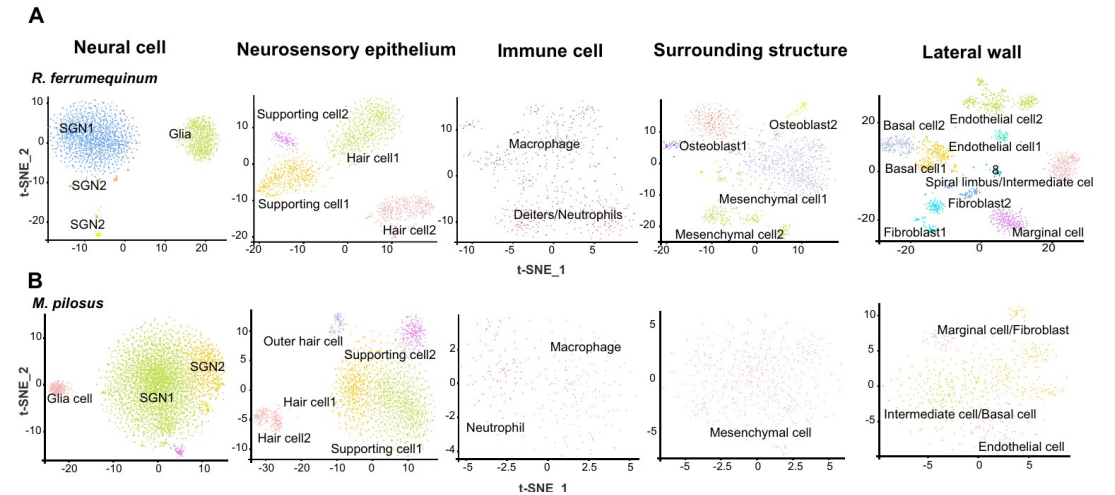
Wang et al. (2025), Fig. 1(B–C). Cell clusters and original marker-expression matrix.

Spiral ganglion neurons carry peripheral auditory signals

Spiral ganglion neurons, or SGNs, are cochlear sensory neurons that convey auditory information toward the brainstem. Wang and colleagues distinguished SGN1 and SGN2 expression categories in both bats. These labels refer to molecular populations in this analysis rather than the onset and offset response categories measured by Suga.

The neural grouping was supported by expression of genes including CELF4 and GABBR2. Glial cells were identified using markers including MBP and PLP1. The same analysis separated hair cells and supporting cells from neural populations, allowing species differences to be assigned to particular tissue compartments.

The atlas found two principal SGN categories, but did not record individual SGN tuning curves. A proposed association between a molecular category and narrowly tuned CF processing therefore remains an interpretation. RNA identity must be linked to recordings before a specific cell population can be assigned an established echo-frequency computation.



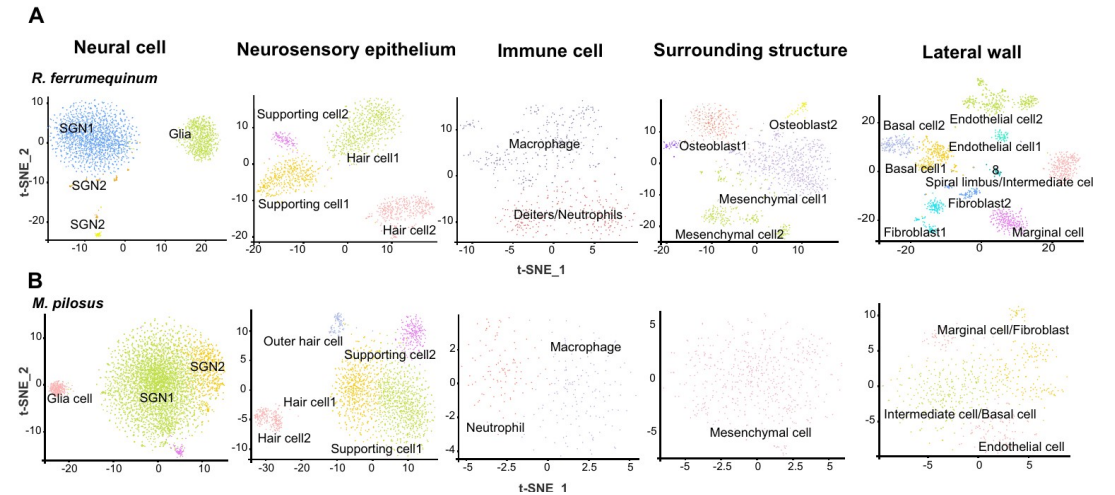
Wang et al. (2025), Fig. 2(A–B). Classified neural and sensory populations.

Hair-cell markers locate candidate channel mechanisms

A **hair cell** is a sensory epithelial cell whose mechanically sensitive structures respond to displacement. In the atlas, expression of CDC14A, XIRP2, Kcnq4 and LBH helped identify hair-cell populations. **KCNQ4** names a potassium-channel gene; detecting its RNA identifies a candidate cellular mechanism rather than measuring a potassium current.

Wang and colleagues separated sensory epithelial populations from spiral ganglion neurons and cochlear lateral-wall cells. Their 14-category horseshoe-bat atlas therefore places a channel-associated marker in a tissue context. Supporting-cell markers included SLC1A3, SPARCL1 and ETV4, emphasizing that sensory transduction depends on a multicellular environment.

The study did not selectively block KCNQ4, measure channel gating or test compensation after a channel intervention. It cannot establish that KCNQ4 alone creates the auditory fovea. The molecular evidence complements peripheral recordings while preserving the distinction between expressed genes, functional membrane currents and behavior.



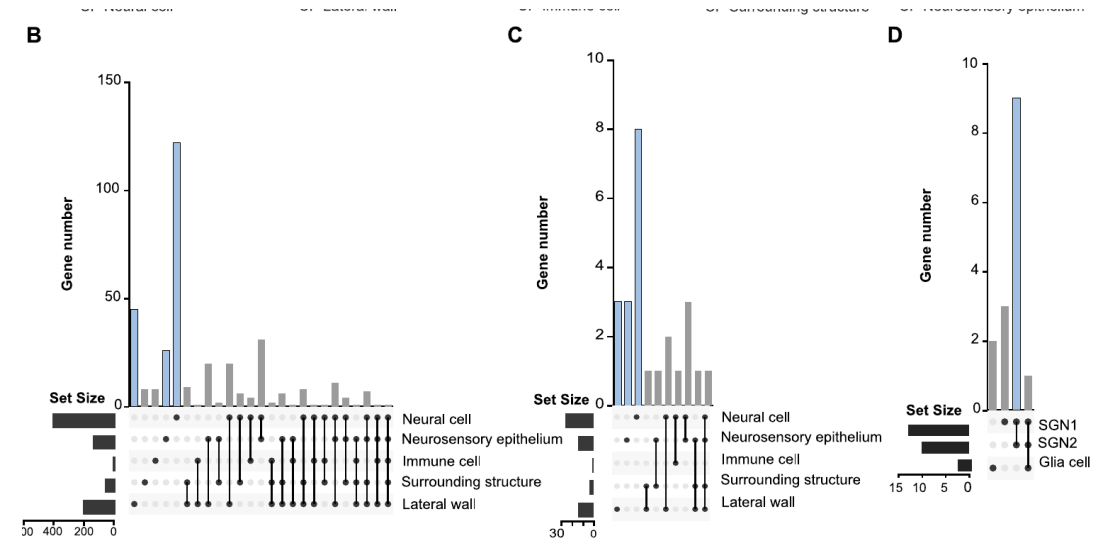
Wang et al. (2025), Fig. 2(A–B). Sensory and supporting-cell populations.

CF-associated expression is concentrated in neural cells

Differential expression describes a difference in measured gene abundance between comparison groups. Wang and colleagues mapped previously identified CF-associated gene sets onto their cochlear cell atlas. The mapped nervous-system-related genes were concentrated in neural cells, particularly the two spiral-ganglion categories.

The analysis localized a broader set of 1365 upregulated genes across neural cells, sensory epithelium and lateral-wall cells. It combined whole-tissue expression comparisons with single-nucleus localization, separating a cochlear-wide molecular signal into candidate cellular sources. These are expression measurements rather than recordings of acoustic responses.

Enrichment in SGN1 and SGN2 supports focused hypotheses about peripheral neural specialization. It does not establish that every mapped gene is necessary for CF hearing, or that its expression changes during compensation. Species differences, tissue preparation and cell classification remain relevant when connecting molecular patterns to behavior.



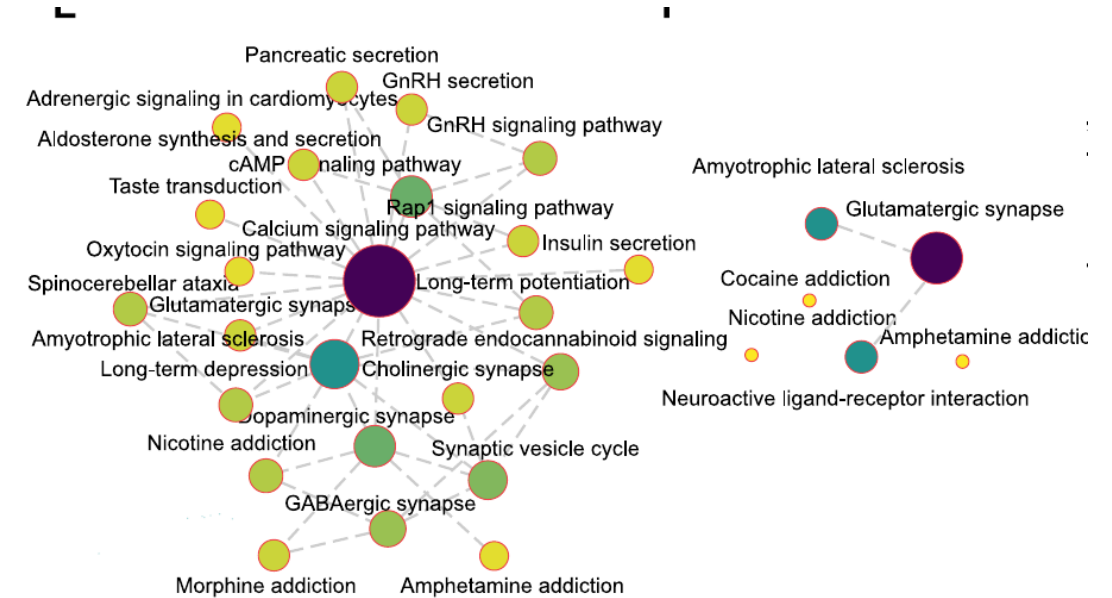
Wang et al. (2025), Fig. 3(B–D). CF-associated genes across cell classes.

Calcium pathways are candidate components of CF tuning

Pathway enrichment asks whether a measured gene set contains many genes assigned to a biological process. In Wang and colleagues' spiral-ganglion analysis, calcium signaling occupied a central position in the horseshoe-bat pathway network. The FM-bat comparison instead emphasized a glutamatergic-synapse pathway among its identified neural categories.

Calcium signaling refers to cellular processes regulated by calcium availability and calcium-sensitive proteins. A **glutamatergic synapse** uses glutamate-associated signaling between cells. The networks were constructed from gene-expression categories; their connections indicate relationships among annotated pathways, not demonstrated synaptic wiring.

The difference provides a molecular candidate for investigating CF-associated auditory specialization. No calcium concentration, channel current or transmitter-release rate was measured in this experiment. A pathway assignment therefore cannot be converted into an established sequence in which one channel changes SGN firing and thereby produces Doppler-shift compensation.



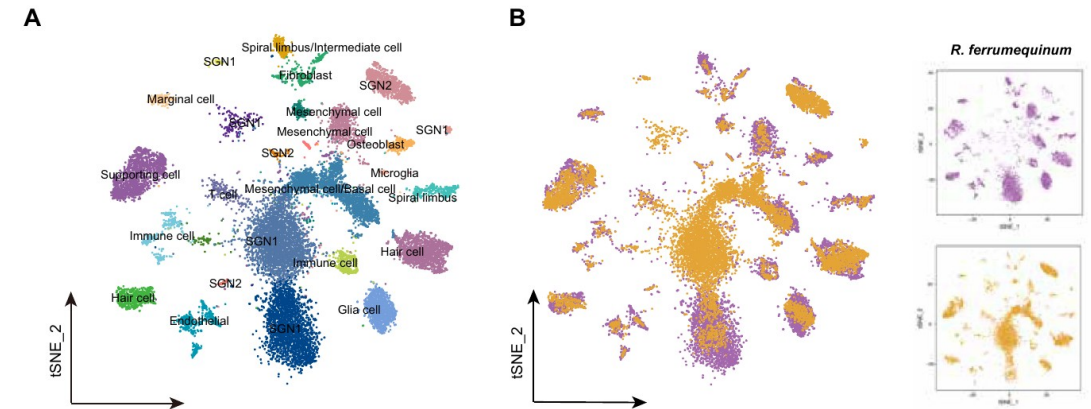
Wang et al. (2025), Fig. 3(E–F). Published pathway networks from cochlear neural cells.

Orthologous genes permit a controlled species comparison

Orthologous genes are corresponding genes in different species inherited from a common ancestral gene. Wang and colleagues used 8607 orthologous genes to construct a joint cochlear atlas for *Rhinolophus ferrumequinum* and *Myotis pilosus*. This provides a common comparison space rather than comparing unrelated species-specific marker lists.

The joint analysis resolved 26 expression clusters and assigned familiar cochlear identities, including SGNs, glial cells, supporting cells and hair cells. Some clusters were dominated by one species, while others contained contributions from both. The separation preserves both shared cellular organization and species-associated expression differences.

A shared gene framework improves comparability but does not isolate echolocation mode from every other species difference. These two species also differ in ancestry and physiology. A CF-associated expression pattern is therefore a candidate adaptation, requiring wider comparative sampling or direct manipulation before its contribution to hearing can be established.



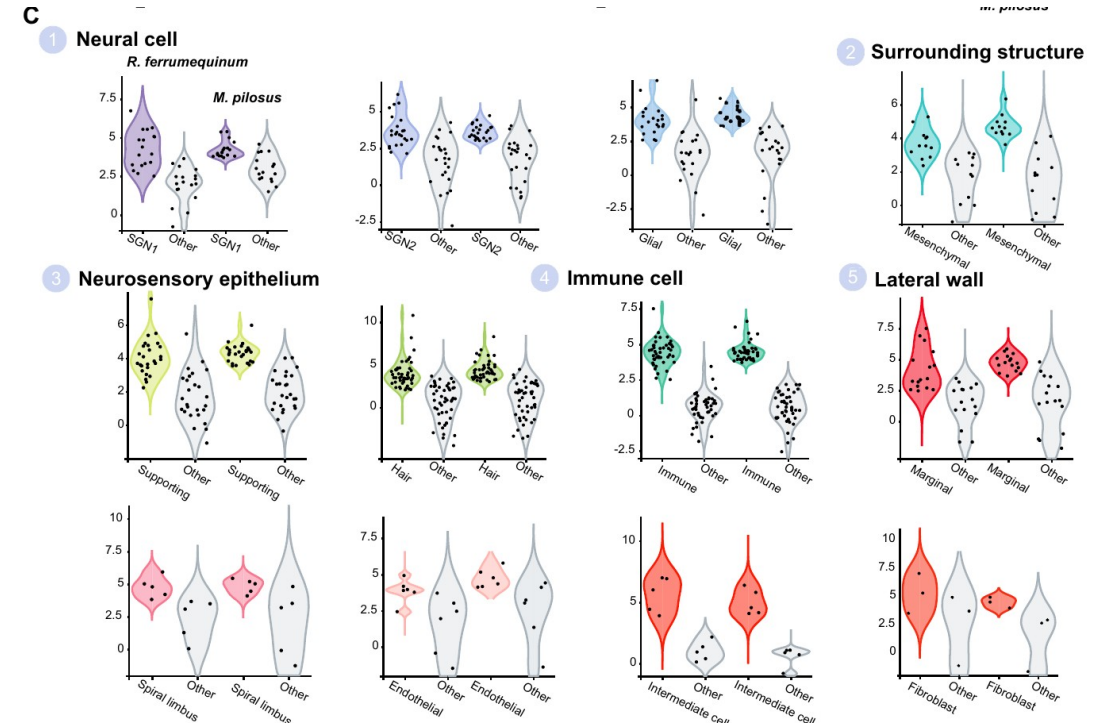
Wang et al. (2025), Fig. 4(A–B). Joint cell atlas and species distribution.

Neural expression categories differ between CF and FM bats

Wang and colleagues compared corresponding cochlear cell types using their joint 26-cluster atlas. Spiral-ganglion populations shared neuronal expression features, while species-associated differences appeared within those broader identities. The distinction concerns the molecular organization of peripheral auditory cells rather than the existence of an entirely separate sensory organ.

Gene-expression distributions describe how RNA abundance varies across a cellular population. The published comparisons preserved cell-type identity while evaluating expression across the CF and FM species. This makes it possible to ask whether a neural category is conserved and whether its molecular profile differs between echolocation strategies.

Molecular heterogeneity does not directly measure acoustic bandwidth. A relatively uniform RNA population could still contain physiologically diverse neurons, and similar RNA profiles need not guarantee identical tuning. The study supports targeted electrophysiological tests rather than assigning a measured kilohertz range to an expression cluster.



Wang et al. (2025), Fig. 4(C). Cross-species expression distributions in identified cell types.

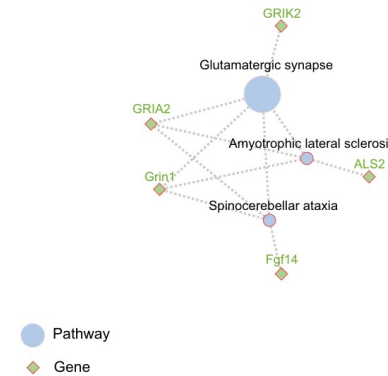
GABA receptor pathways suggest testable inhibitory roles

GABA is an inhibitory transmitter in many neural systems. Wang and colleagues identified GABAergic-synapse pathway enrichment in a horseshoe-bat SGN1 category, alongside glutamatergic, dopaminergic pathways associated with the transmitter dopamine, and calcium-signaling pathways. Their analysis associated 233 highly expressed genes with this species-specific neuronal category.

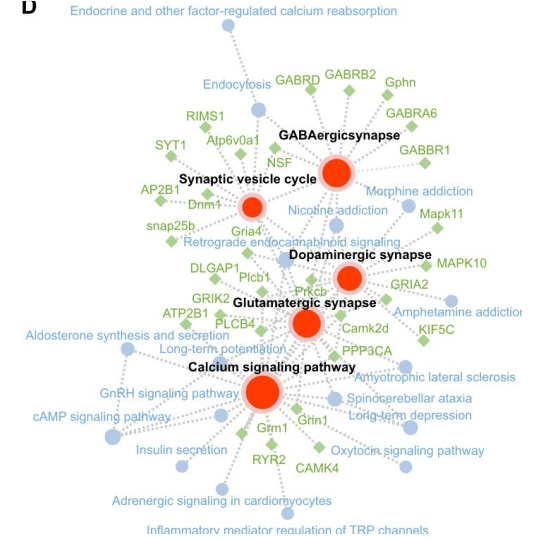
The pathway included GABA A and GABA B receptor annotations. These name distinct receptor classes through which GABA can influence cellular signaling. The atlas measured RNA, not receptor binding, synaptic inhibition or a response to a pharmacological blocker. Its two receptor annotations therefore identify candidates rather than a demonstrated inhibitory circuit.

The authors proposed a connection to specialized CF processing and vocal compensation. That proposal does not establish that cochlear GABA receptors perform the auditory comparison controlling call frequency. A causal account requires receptor-specific interventions, neuronal measurements and a behavioral compensation assay in the same mechanistic chain.

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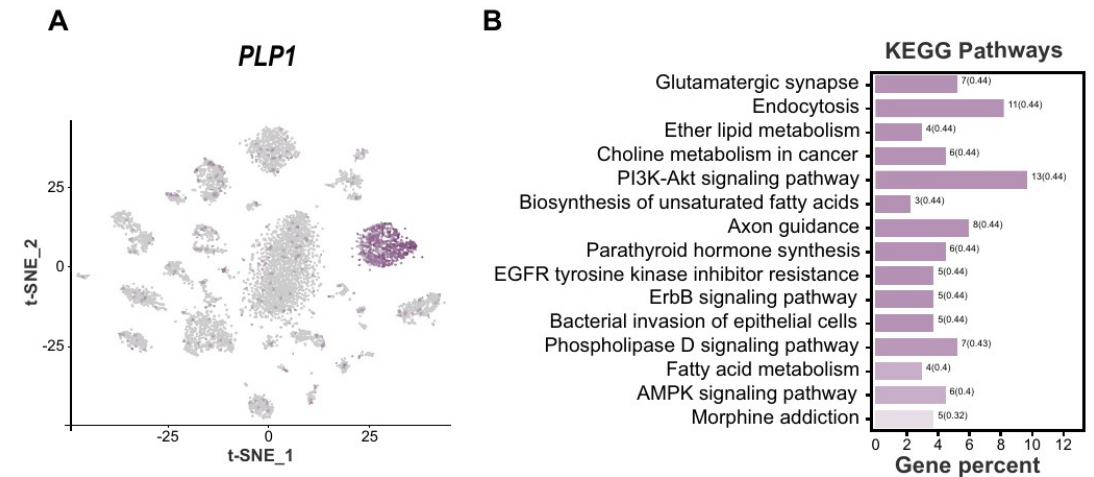
Wang et al. (2025), Fig. 5(C–D). Original neuronal expression and pathway-enrichment panels.

Glial signaling is inferred from expression relationships

Glial cells support and interact with neurons without being classified as the principal signal-carrying sensory neurons. Wang and colleagues localized PLP1 expression in their cochlear atlas and examined highly expressed genes within the glial category. The analysis associated this category with cell communication and several neural-support processes.

The horseshoe-bat atlas distinguishes glia from SGN1 and SGN2, allowing support-cell expression to be studied separately from neuronal expression. **Ligand–receptor inference** predicts possible communication when one cell class expresses a signaling molecule and another expresses a corresponding receptor. Such inference is based on molecular compatibility, not direct observation of transmission.

The study proposed interactions involving neural adhesion and growth-associated signaling. It did not stimulate glia, measure a resulting SGN membrane response or test an alteration of echo compensation. Glial molecular specialization remains a candidate contributor to the sensory environment rather than an established moment-to-moment Doppler controller.



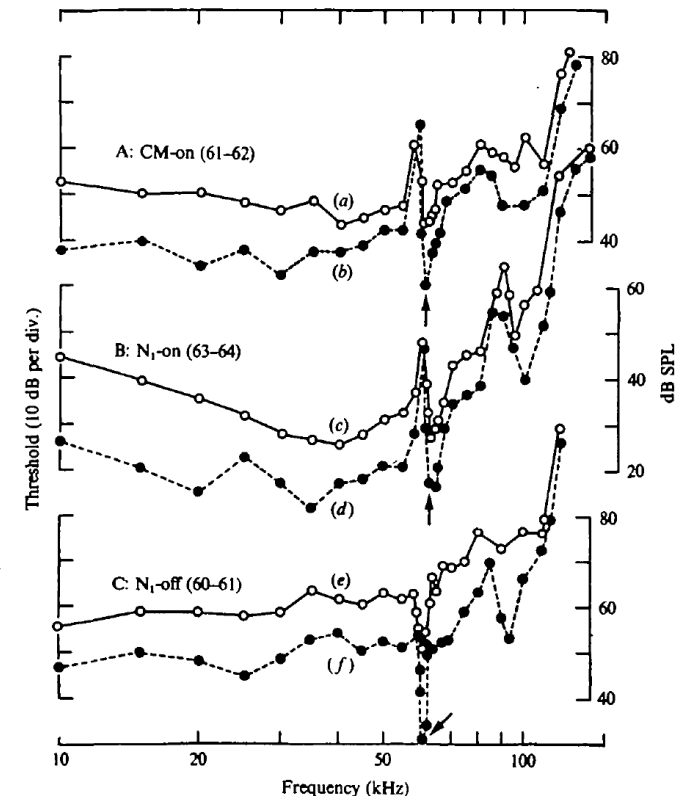
Wang et al. (2025), Fig. 6(A–B). Glial marker localization and functional gene categories.

The cochlear response is sharply tuned near CF2

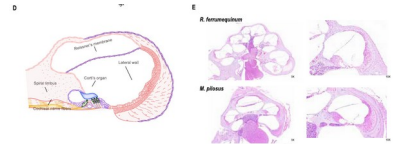
CF2 is the constant-frequency component of the second harmonic. In Suga and colleagues' mustached bats, *Pteronotus parnellii*, CF2 lay near 61–62 kHz. The CM threshold curve contained a pronounced sensitivity notch in approximately this band.

The best frequencies depended on the recorded response: CM onset was most sensitive near 61–62 kHz, summed neural onset near 63–64 kHz, and summed neural offset near 60–61 kHz. **Onset** and **offset** refer to the beginning and cessation of the stimulus.

A sharply tuned receptor-related response indicates specialization before cortical processing. The difference between neural onset and other thresholds requires attention to the response envelope and synchrony; it does not mean the hearing system's preferred echo band is simply 63–64 kHz.



Suga et al. (1975), Fig. 2(A–C). Threshold notches of CM onset, neural onset, and neural offset, in kHz.



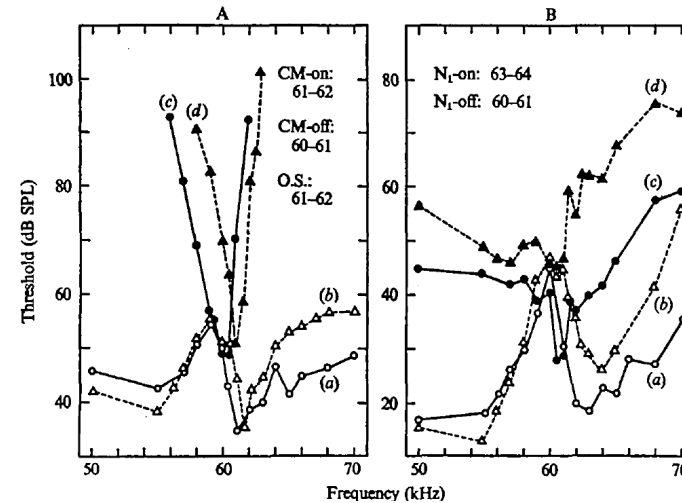
Wang et al. (2025), Fig. 1(D–E). Published cochlear anatomy and stained tissue sections.

A cochlear resonator prolongs excitation after sound ends

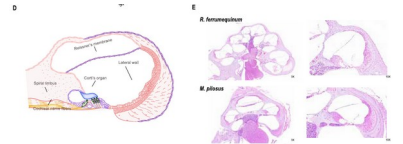
A **resonator** responds especially strongly near a particular frequency and can continue oscillating after stimulation stops. Near the CF2 band, Suga and colleagues found that CM activity built up and decayed slowly compared with responses farther from the tuned range.

The residual CM activity after stimulus cessation was called **CM-after**. Its transient behavior was consistent with mechanical ringing in the inner ear. A neural offset response accompanied these peripheral events rather than requiring a new acoustic pulse.

Resonance links high frequency selectivity to a distinctive time course. The ear does not reproduce the imposed tone-burst envelope identically at every frequency. A sound with a sharp external ending can therefore produce continuing or changing receptor excitation inside the cochlea.



Suga et al. (1975), Fig. 3(A-B). Different best frequencies of cochlear and neural onset and offset responses.



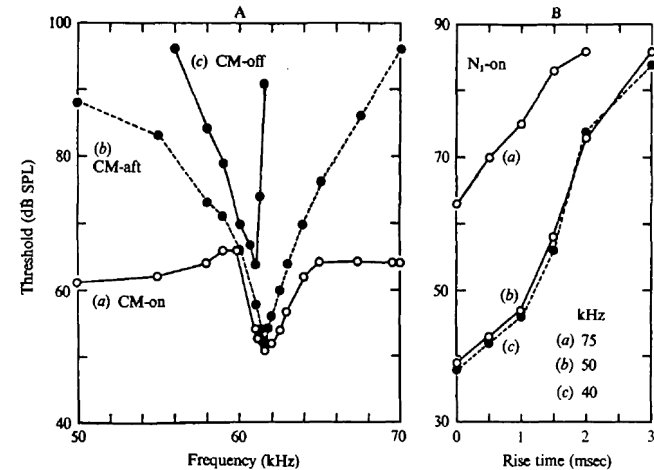
Wang et al. (2025), Fig. 1(D-E). Published cochlear anatomy and stained tissue sections.

Onset synchrony depends on the sound envelope

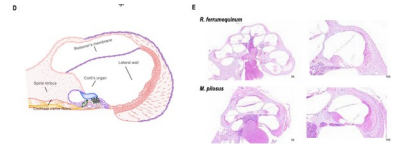
The **envelope** is the changing amplitude of a sound or response over time. Slow cochlear build-up near resonance can spread the onset times of individual neural impulses. A summed neural onset response becomes smaller when those impulses are less synchronized.

Suga and colleagues varied the rise time of tones away from the resonant band. Lengthening rise time raised the threshold of the summed onset response. A 1 ms rise produced an approximately 8 dB threshold increase relative to an abrupt rise; a 3 ms rise produced a much larger increase.

This control supports an envelope-based explanation of the apparent onset-threshold shift. A weak summed response can result from dispersed spike timing even when individual neurons remain sensitive. Population synchrony and acoustic sensitivity must be distinguished.



Suga et al. (1975), Fig. 4(A-B). Cochlear transients and the dependence of neural onset threshold on rise time.



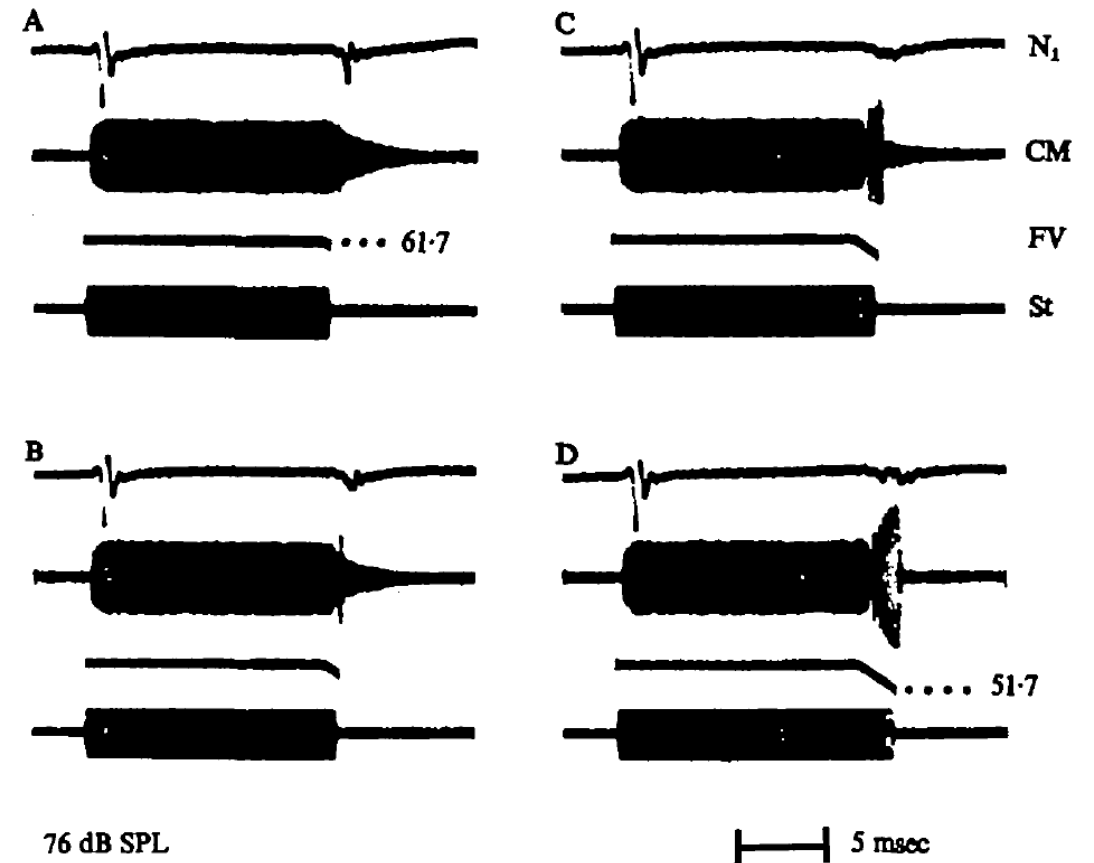
Wang et al. (2025), Fig. 1(D-E). Published cochlear anatomy and stained tissue sections.

Terminal FM suppresses cochlear offset transients

Suga and colleagues compared a CF tone ending abruptly with the same tone followed by a downward FM component. The added sweep changed the cochlear response after the sustained tone rather than merely adding a second independent response.

A terminal FM sweep reduced residual CM activity and the summed neural offset response under the tested conditions. The effect depended on the sweep's extent and duration. CF and FM therefore interact in the peripheral time course of excitation.

The finding connects the acoustic structure of a natural CF-FM signal to cochlear mechanics. It does not establish that every terminal sweep eliminates all residual excitation during free flight. The stimulus comparisons isolate how a controlled FM ending modifies a tuned CF response.



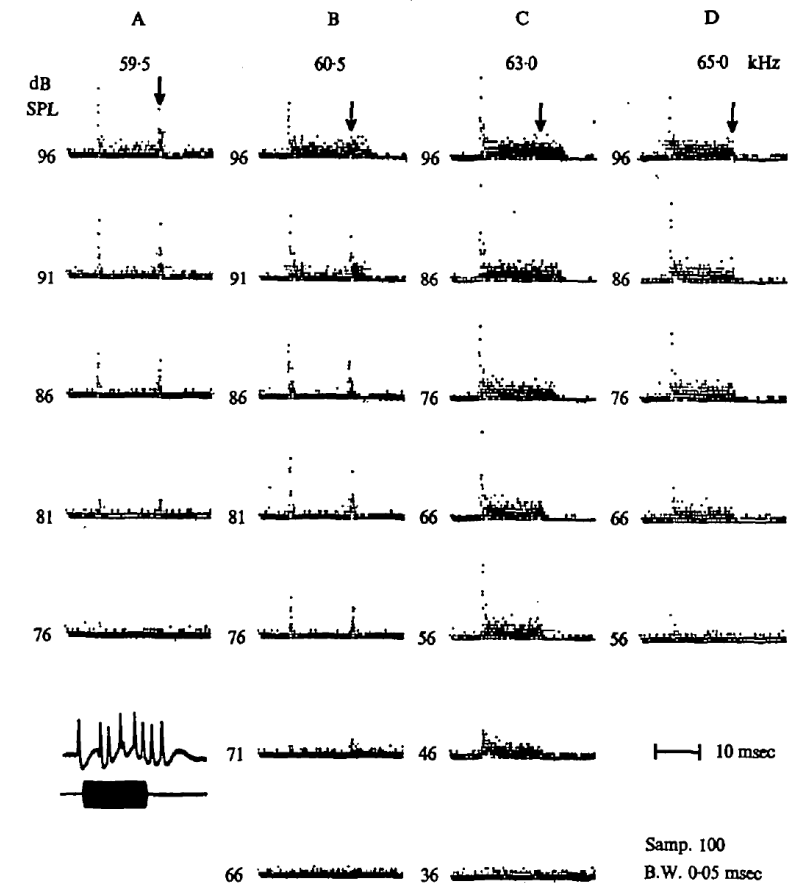
Suga et al. (1975), Fig. 8(A-D). Terminal FM changes cochlear ringing and the summed neural offset response.

Peripheral neurons can discharge when a tone ends

Suga and colleagues recorded individual peripheral auditory neurons, sampling the auditory nerve and nearby cochlear nucleus. A **post-stimulus-time histogram** aligns impulses from repeated trials to the onset of the same stimulus.

Individual neurons could produce sustained activity during a tone and concentrated discharges after its cessation. For some tested frequencies and levels, the offset discharge remained prominent even when sustained activity was weak.

These offset responses linked the summed N1-offset signal to individual neural activity. They occurred in neurons with excitatory tone responses, rather than requiring a separate population responding only at offset. Peripheral sampling and recording location limit claims about an exclusively cortical mechanism.



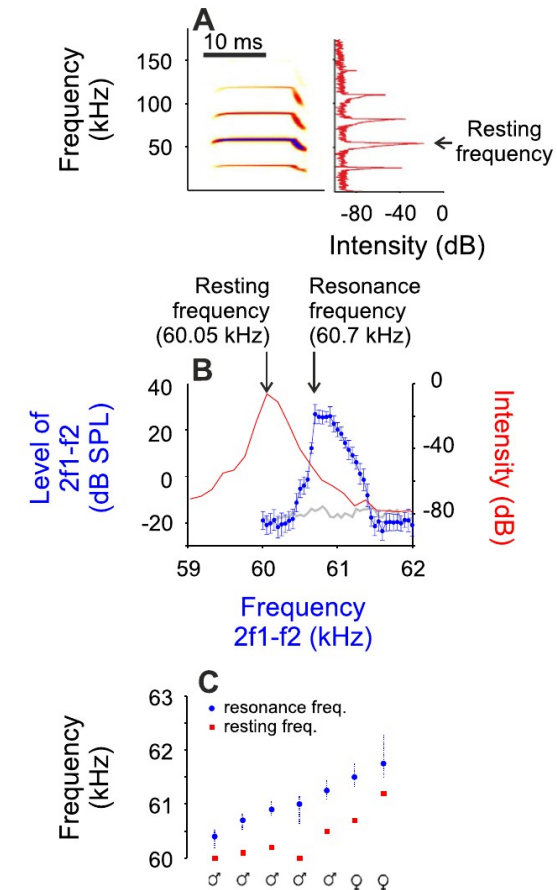
Suga et al. (1975), Fig. 9(A-D). Single-neuron responses across tone frequency and level, with offset discharges.

Resonance frequency differs from the resting call baseline

Resonance frequency is the frequency at which the peripheral resonant system responds especially strongly. Macías and colleagues distinguished it from the resting second-harmonic call frequency in mustached bats. Resting emissions ranged from 60 to 61.2 kHz in their animals, while the resonance estimate generally lay above the emitted baseline.

They combined acoustic recordings with auditory-midbrain recordings and examined sharply tuned neurons near the specialized band. A reference defined from vocal emission therefore need not coincide exactly with a reference defined from the sensory response. This distinction parallels the resting-versus-reference frequency difference in behavioral compensation experiments.

The measurement distinguishes two biologically related frequency values rather than assigning one universal CF frequency to the species. Neural tuning was evaluated relative to the individual bat's acoustic and peripheral specialization. It does not follow that every midbrain cell near this band contributes directly to the vocal feedback pathway.



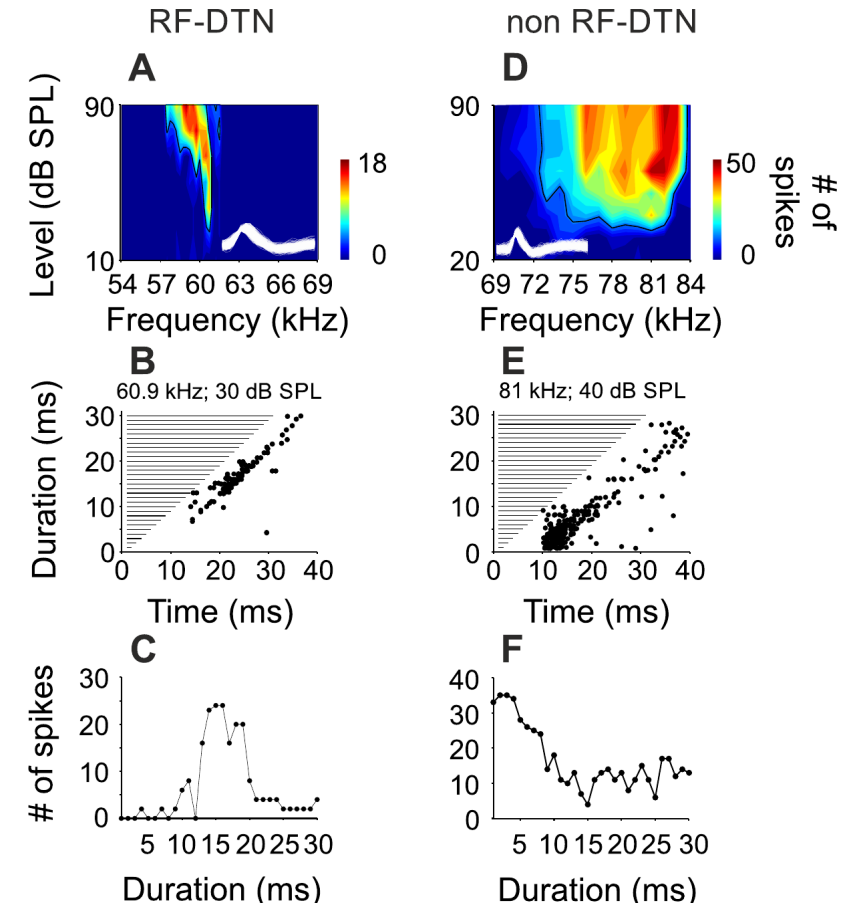
Macías et al. (2016), Fig. 1(A–C). Resting emissions and cochlear resonance estimates.

Midbrain neurons can prefer a narrow sound duration

A **duration-tuned neuron** responds most strongly to a restricted range of sound durations. Macías and colleagues recorded single-neuron activity in the mustached bat's **inferior colliculus**, an auditory midbrain structure. Pure-tone stimuli varied in frequency, sound level and duration, separating several dimensions of response selectivity.

One resonance-band neuron preferred 60.9 kHz, with a frequency bandwidth of 0.75 kHz. It preferred a 15-ms stimulus, with a duration bandwidth of 7 ms. A comparison neuron tuned to 81 kHz had a broader 9-kHz frequency bandwidth and preferred a 3-ms stimulus.

These examples establish joint spectral and temporal selectivity in central auditory neurons. They do not measure the neuron's synaptic inputs or identify its transmitter. Sharp duration tuning can coexist with sharply tuned peripheral input, so long cochlear ringing does not force every central response to be temporally broad.



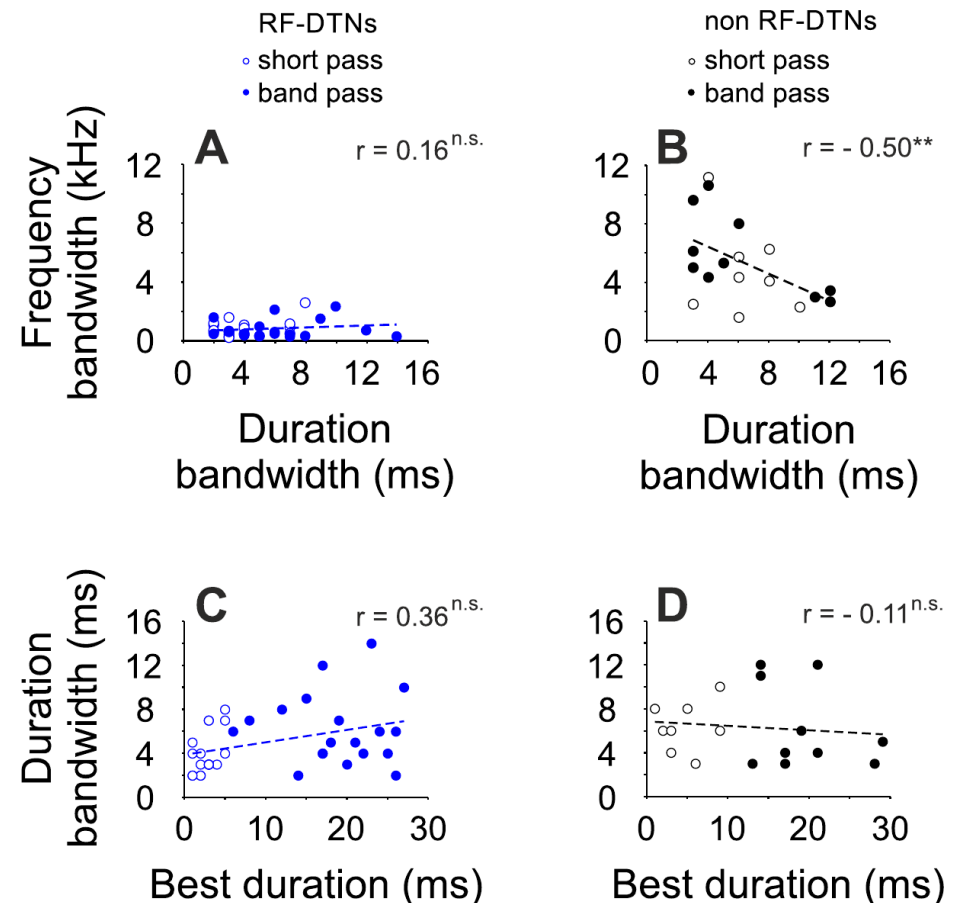
Macías et al. (2016), Fig. 2(A–F). Frequency response areas and measured duration tuning.

Sharp spectral tuning need not sacrifice duration tuning

A **spectral–temporal trade-off** occurs when finer frequency resolution accompanies poorer temporal resolution. Cochlear resonance can sustain activity after sound offset, extending the peripheral response. Macías and colleagues asked whether this peripheral property constrains duration-selective neurons in the auditory midbrain.

They measured frequency bandwidth at 10 dB above threshold and duration bandwidth across stimulus durations. Resonance-band neurons did not follow the expected relation between narrow frequency tuning and broad duration tuning. Neurons outside that specialized band displayed the more conventional relationship between the two response dimensions.

The result concerns central response selectivity under controlled pure-tone stimulation. It does not mean cochlear ringing disappeared or that every auditory neuron overcomes the trade-off. Instead, central processing can produce sharp temporal filtering despite a peripheral input specialized for narrowband frequency analysis.



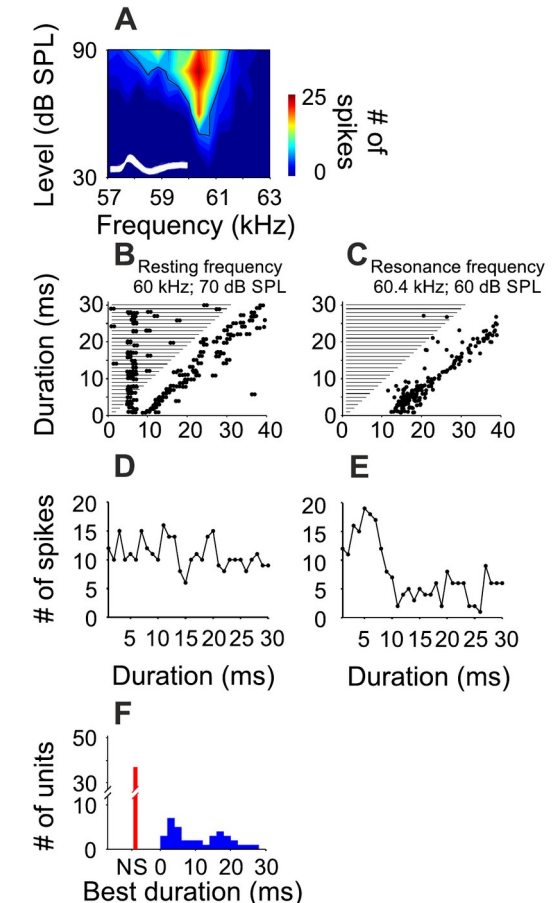
Macías et al. (2016), Fig. 4(A–D). Relations between frequency and duration bandwidths.

A small frequency change can reorganize duration tuning

Macías and colleagues tested the same resonance-band neurons with tones at the individual resting-call frequency and at the resonance frequency. This within-neuron comparison controlled cell identity while changing the acoustic frequency. The resting frequency lay just below the specialized resonant band.

Duration selectivity was clear at resonance but often absent at the resting frequency. Preferred durations at resonance clustered near 4 ms and 17 ms. The response to resting-frequency stimulation could instead remain broad across durations, despite the small spectral separation between the two stimulus conditions.

Frequency-dependent temporal filtering means that a neuron's response to duration depends on the carrier frequency. It cannot be described adequately by one duration preference measured at any arbitrary tone frequency. The experiment supports interaction between spectral input and central temporal processing without directly identifying the synapses responsible.



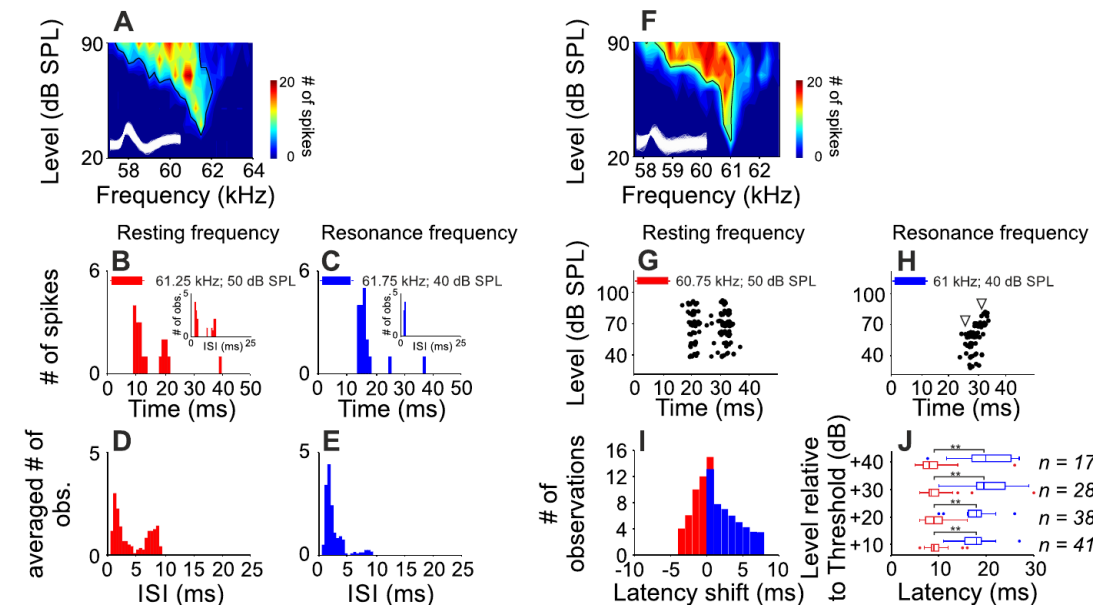
Macías et al. (2016), Fig. 5(A–F). Responses at resting and resonance frequencies.

Resonance-band inhibition is still a hypothesis

Inhibition reduces neuronal output through synaptic or membrane mechanisms. Macías and colleagues proposed frequency-specific inhibition to explain why resonance-band neurons could be selective for duration despite cochlear ringing. Their recordings measured output spikes under different stimulus frequencies and sound levels.

The published response patterns and latency changes differed between resting and resonance stimulation. Duration tuning could be present at resonance but absent just below it. The authors considered temporally offset excitation and inhibition, including coincidence and anti-coincidence accounts, as explanations for this frequency-dependent selectivity.

No inhibitory transmitter was blocked in this study, and no inhibitory synaptic current was directly recorded. The evidence therefore supports a central filtering hypothesis rather than proving a particular GABA receptor or channel mechanism. Behavioral compensation and central duration tuning also remain separate measured phenomena, even though both depend on the specialized CF band.



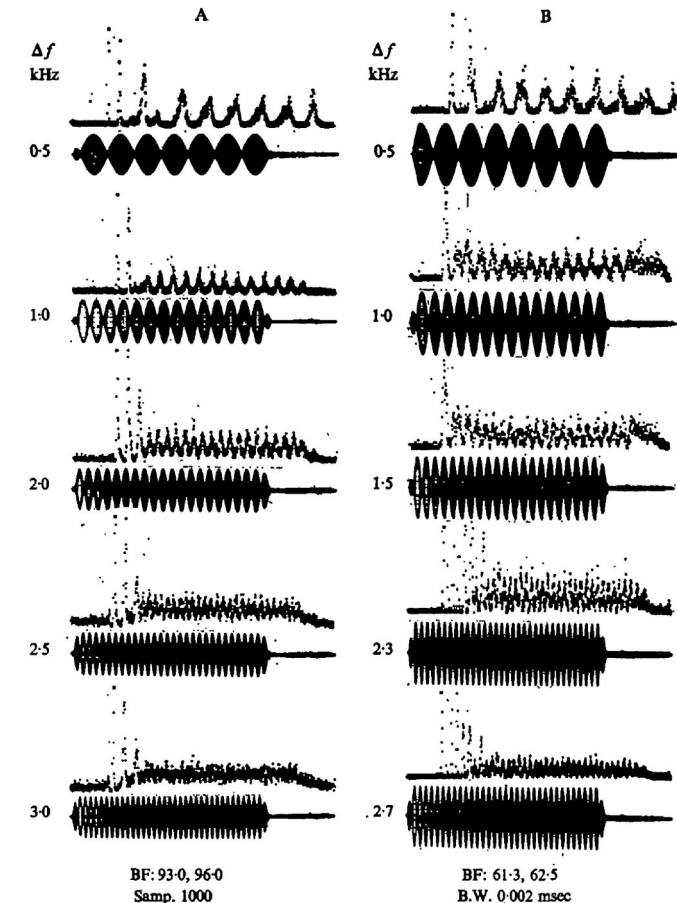
Macías et al. (2016), Fig. 6(A–J). Frequency-dependent response timing and latency shifts.

Peripheral impulses can track beats between nearby tones

Beats are periodic amplitude fluctuations produced when tones of slightly different frequency overlap. Suga and colleagues presented paired tones and recorded whether peripheral impulses followed the resulting fluctuations.

Phase locking means that discharge occurs at a consistent phase of a repeating stimulus feature. Peripheral neurons could phase-lock to beats up to about 3 kHz. Changing the separation between tones changed the rate of the amplitude fluctuations.

This timing code complements the code supplied by sharply tuned best frequencies. Overlapping emitted CF signals and returning echoes can provide beat information, but the paired-tone recordings do not establish which cue a freely flying bat relies on in every compensation decision.



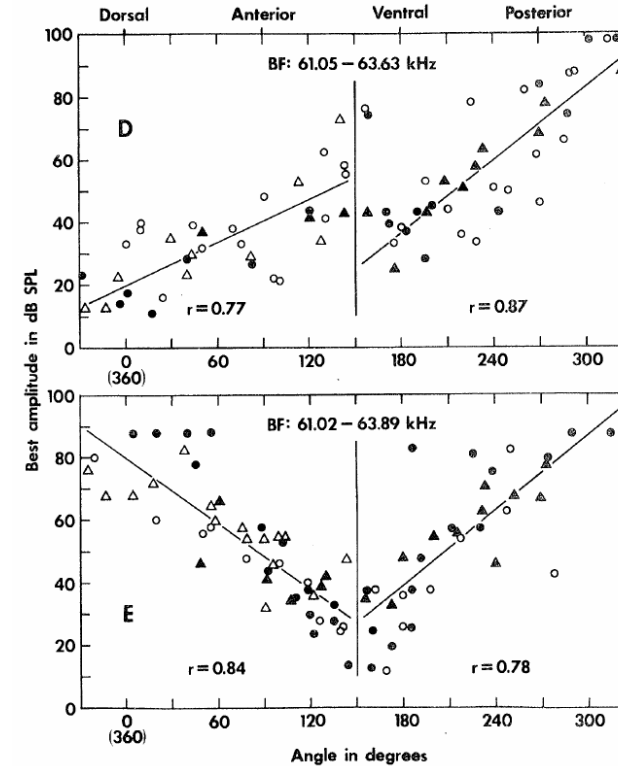
Suga et al. (1975), Fig. 18(A-B). Peripheral impulse timing follows beat envelopes at different frequency separations.

The cortex devotes extensive space to the CF echo band

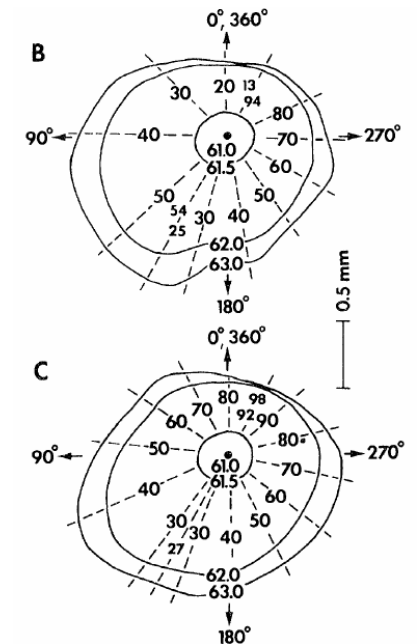
The **primary auditory cortex** is a cortical region receiving organized auditory input. In Suga's mustached-bat study, the Doppler-shifted-CF processing area occupied about 2.3 mm², approximately 30% of primary auditory cortex.

Its neurons were concentrated near the 61–63 kHz band associated with CF components of emitted signals and echoes. This large representation mirrors the peripheral specialization but adds cortical response properties beyond a neuron's best frequency.

Tonotopy is an orderly representation of frequency across neural tissue. Suga examined how neurons sharing a best frequency differed in preferred sound amplitude. The enlarged CF region was not simply a collection of interchangeable neurons carrying identical information.



Suga (1977), Fig. 2(D-E). Measured best amplitude varies with cortical position, in dB SPL and degrees.



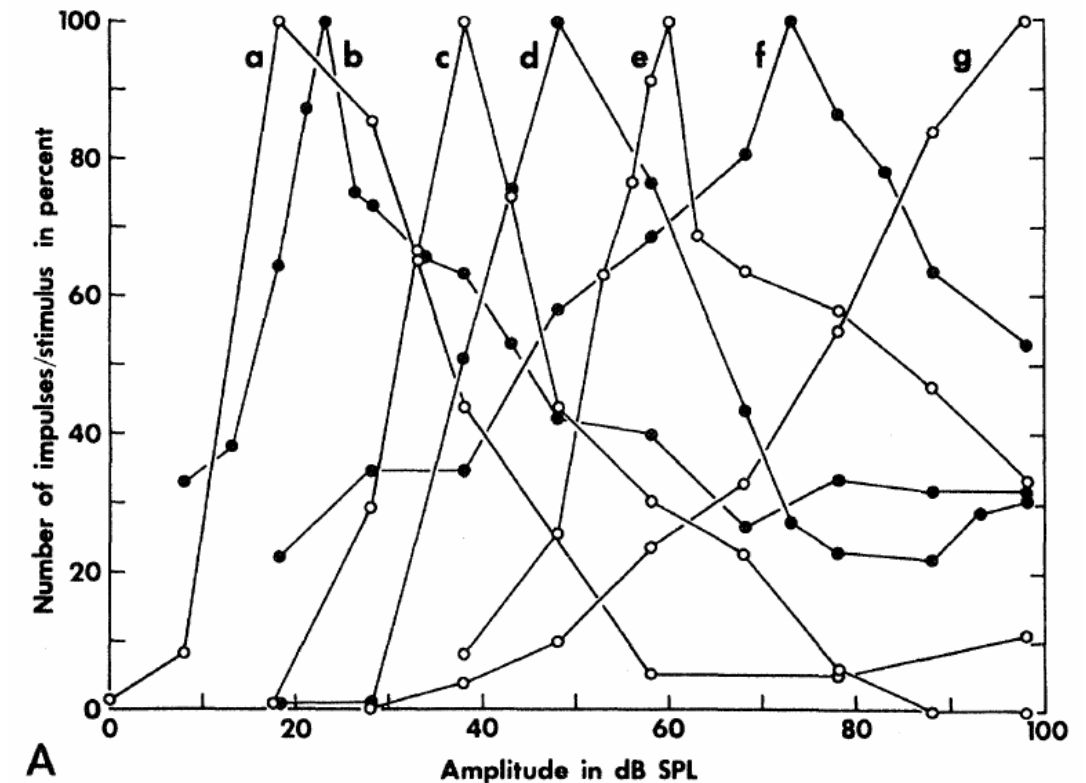
Suga (1977), Fig. 2(B-C). Original cortical maps, 0.5 mm.

Some cortical neurons prefer an intermediate sound level

A neuron's **best amplitude** is the sound pressure level producing its largest response. Suga measured impulse counts during 50 ms tones at each cortical neuron's best frequency and compared response growth across sound levels.

Many CF-region neurons had **nonmonotonic** responses: impulse counts rose with amplitude, reached a maximum, then fell at stronger levels. Other neurons showed more nearly monotonic growth. Preferred amplitude therefore differed from minimum threshold.

An **upper threshold** marks a level above which excitation becomes weak or absent. These cortical response limits create amplitude selectivity that cannot be described solely as increasing discharge to louder sounds. Frequency and sound level jointly determine which cortical populations are most active.



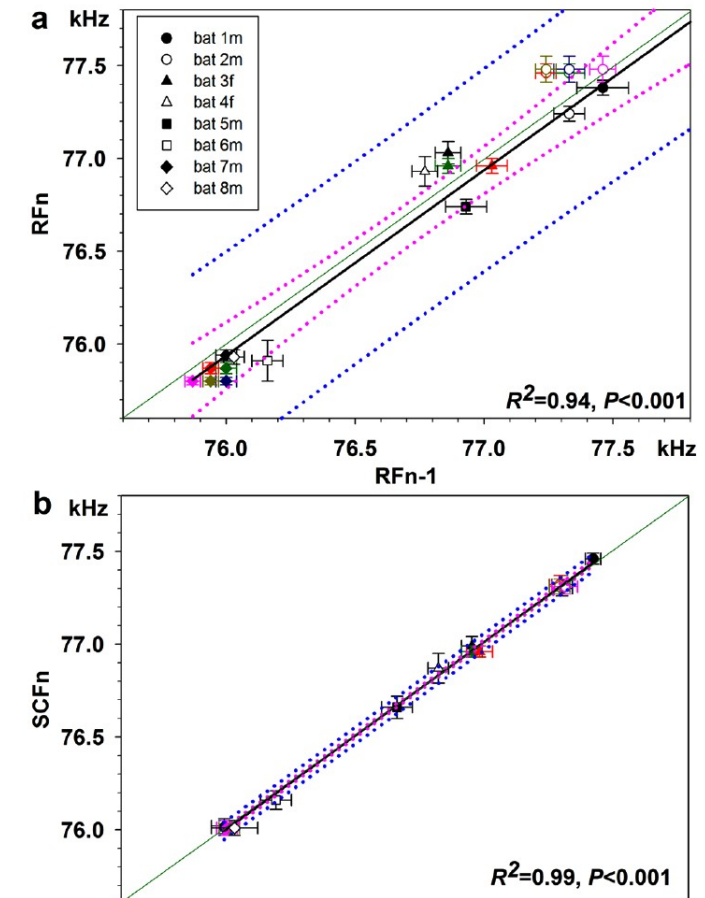
Suga (1977), Fig. 1(A). Cortical impulse-count functions have different preferred sound levels.

Communication calls need not share echolocation feedback

Liu and colleagues compared resting echolocation calls with **short constant-frequency communication calls** in greater horseshoe bats. These social calls resembled shortened CF signals but occurred during interactions between bats. Recording both call types across days tested whether similar acoustic signals shared the same frequency stability.

Resting echolocation frequency varied across days, whereas the communication-call frequency remained much more stable. Typical call durations also differed: resting echolocation calls were approximately 47 ms, while the short social calls were approximately 14 ms. Frequency regulation therefore depended on vocal context rather than acoustic similarity alone.

The authors proposed different feedback control for echolocation and communication, possibly including reduced or absent auditory feedback for the social calls. Their comparison did not directly interrupt feedback during social calling. Stable communication frequency supports that interpretation but does not prove that these calls are produced without auditory input.



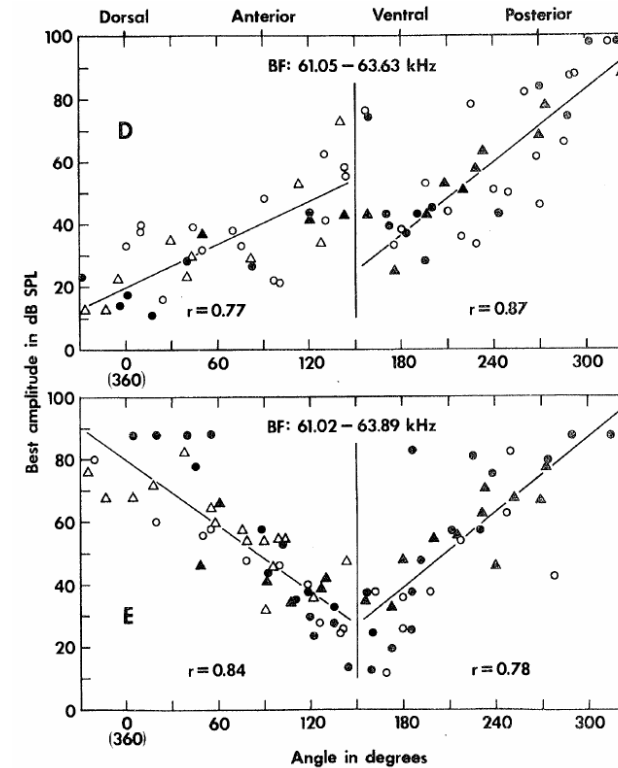
Liu et al. (2013), Fig. 2(A–B). Daily frequency comparisons for echolocation and social calls.

Amplitude and frequency occupy cortical coordinates

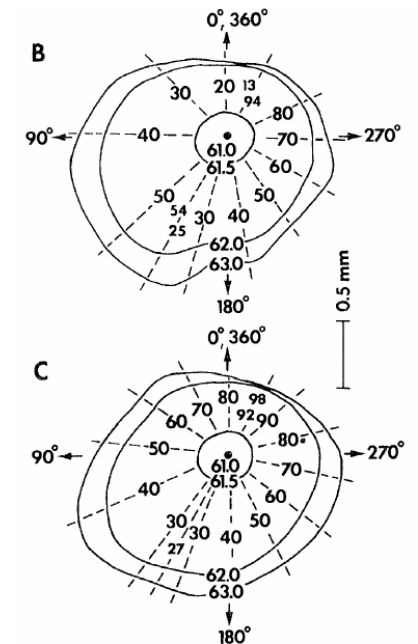
Amplitopic representation is an orderly spatial organization of preferred sound amplitude. In Suga's mapped CF processing area, best-frequency contours and best-amplitude contours followed different directions across the cortical surface.

The best-amplitude organization varied among animals. The posterior region consistently placed larger preferred amplitudes dorsally, whereas the anterior region could follow either of two arrangements. The data support structured amplitude representation without a single identical map in every bat.

Preferred amplitudes of about 30–50 dB SPL occupied more cortical area than preferred amplitudes of about 80–100 dB SPL. This unequal allocation describes the measured map. Its relation to commonly encountered echo levels was an interpretation rather than a direct measurement of natural echo statistics.



Suga (1977), Fig. 2(D–E). Measured best amplitude varies with cortical position, in dB SPL and degrees.



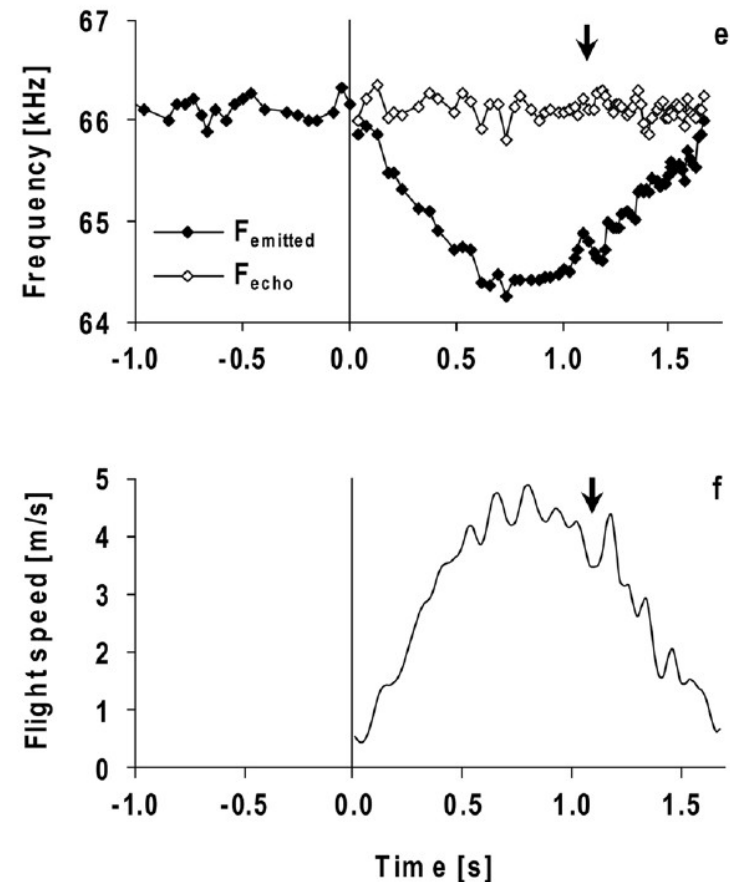
Suga (1977), Fig. 2(B–C). Original cortical maps, 0.5 mm.

Free flight lowers emission while preserving echo frequency

Schoeppler and colleagues studied *Hipposideros armiger* flying toward a landing grid. They recorded echolocation signals, reconstructed the flight trajectory, and calculated the frequency of echoes from targets ahead using the measured flight speed.

During acceleration, emitted CF2 frequency decreased. In a representative flight, the minimum emission frequency coincided with a maximum speed of 4.7 m/s. The calculated returning frequency remained close to the animal's reference band.

This separates the changing outgoing signal from the controlled echo estimate. The study used natural flight, but the displayed echo frequency was calculated from call frequency and kinematics; it was not a direct recording from the bat's cochlea or a microphone at its ear.



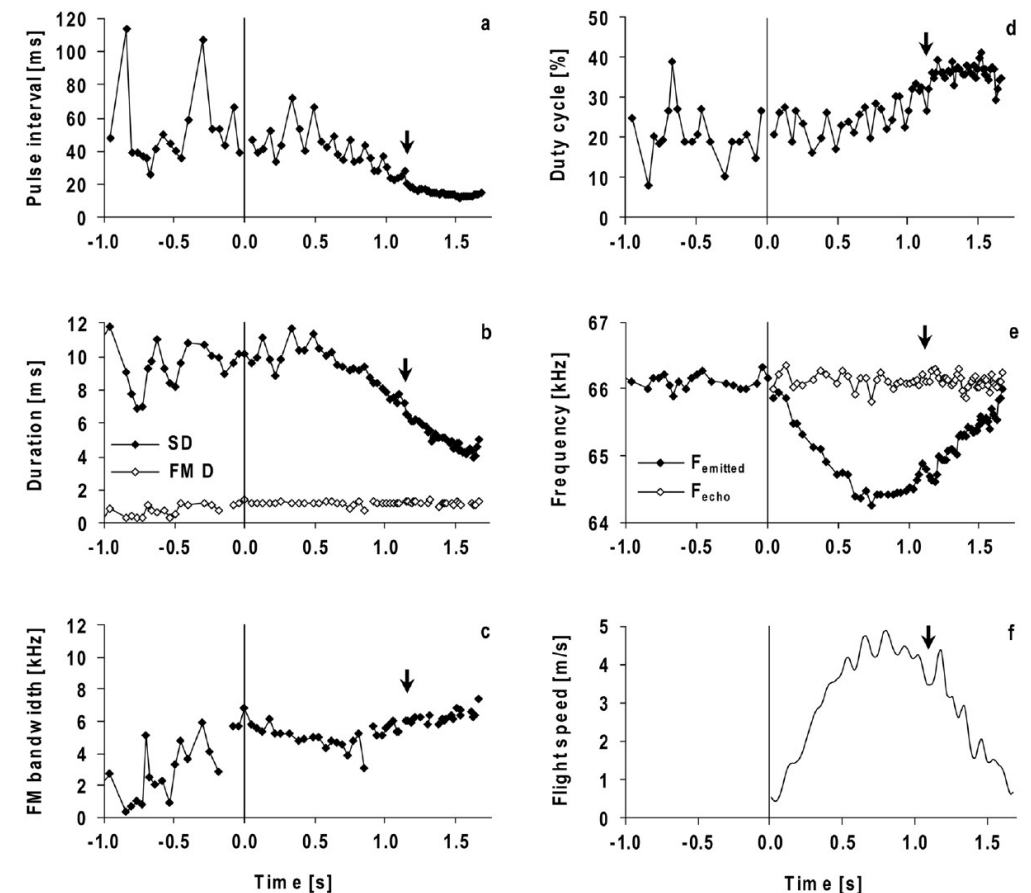
Schoeppler et al. (2018), Fig. 2(e-f). Measured emission and flight speed, with calculated echo frequency.

Approach changes call timing while CF is retained

Pulse interval is the time between consecutive calls;
duty cycle is the proportion of time occupied by sound emission. Both describe how frequently a bat samples its surroundings, independently of CF frequency.

In *H. armiger*, pulse intervals and call durations decreased toward landing. During the final approach, mean call duration reached about 3–4 ms and pulse interval about 11–12 ms. Duty cycle increased, while the CF component remained present.

Frequency compensation therefore operates within an acoustic sequence whose temporal structure is also changing. The terminal approach is not a sequence of otherwise identical calls. Timing, harmonic prominence, and the FM component can change while the returning CF band remains regulated.



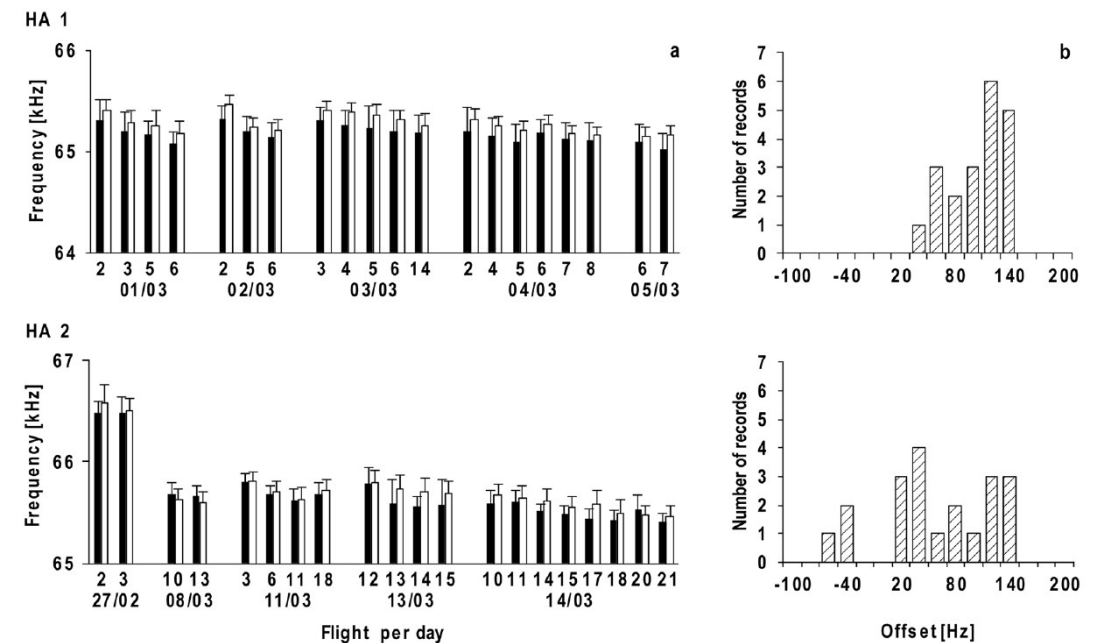
Schoeppler et al. (2018), Fig. 2(a–f). Call timing, FM properties, duty cycle, and compensation during approach.

Resting and reference frequencies shift together

Schoeppler and colleagues measured resting frequency immediately before each flight and reference frequency from the corresponding calculated echoes. The absolute values changed between flights and across recording days, rather than remaining permanently fixed.

The two frequencies remained closely coupled. Their mean offset was about 80 Hz, with the reference usually above the resting frequency. In one bat, reference frequency declined by about 1 kHz over nine days.

A drifting absolute frequency does not by itself imply poor compensation. Comparing an echo estimate from one time with a resting measurement from another can exaggerate the offset. A contemporaneous baseline is required to distinguish movement of the preferred band from error around that band.



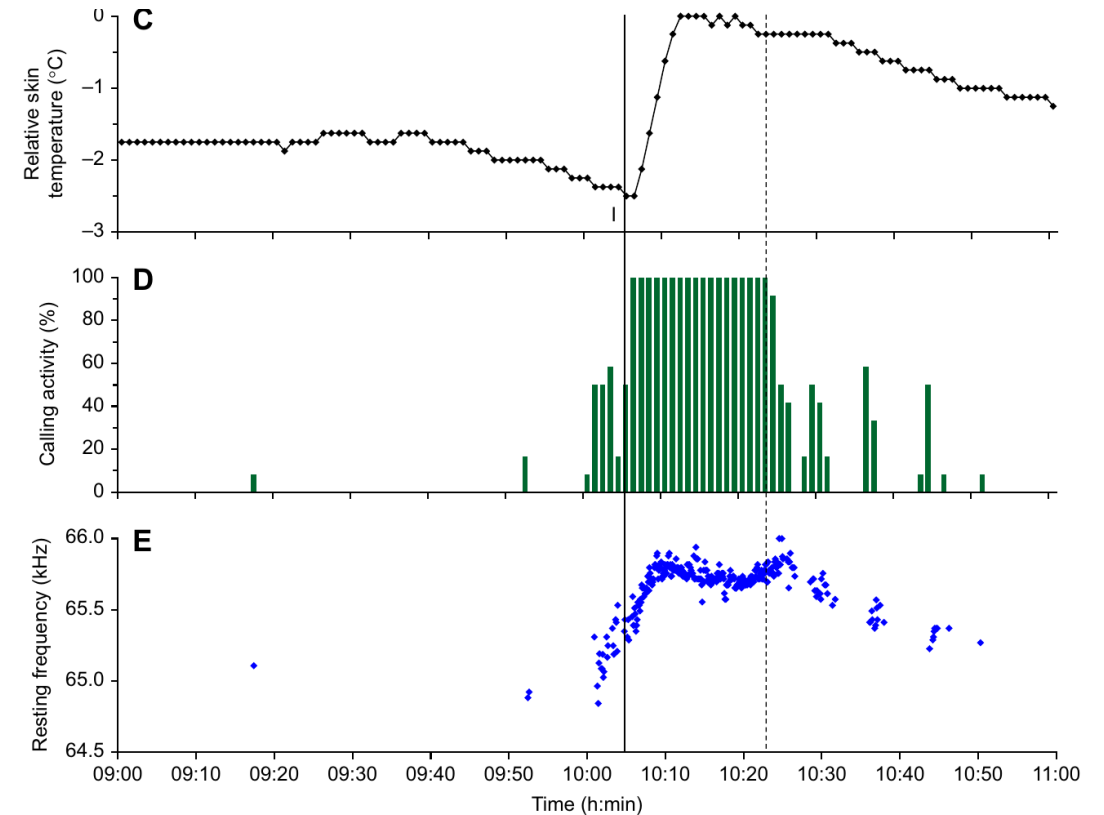
Schoeppler et al. (2018), Fig. 3(a–b). Paired resting and reference frequencies and their offsets across flights.

Resting call frequency rises as torpid bats warm

Torpor is a state of reduced physiological activity associated here with lower temperature. Schoeppler and colleagues attached a miniature logger between the shoulder blades of *H. armiger* and recorded skin temperature together with echolocation calls.

After the experimenter entered the room, resting bats became more active, warmed, and increased their CF2 resting frequency. The largest frequency rise during activation reached about 1.44 kHz. The change occurred without a flight-induced Doppler shift.

The experiment links physiological state to the frequency emitted at rest. It does not isolate temperature from every accompanying change in activity, and the logger measured skin rather than cochlear temperature. The result requires a baseline that can vary even when the animal is stationary.



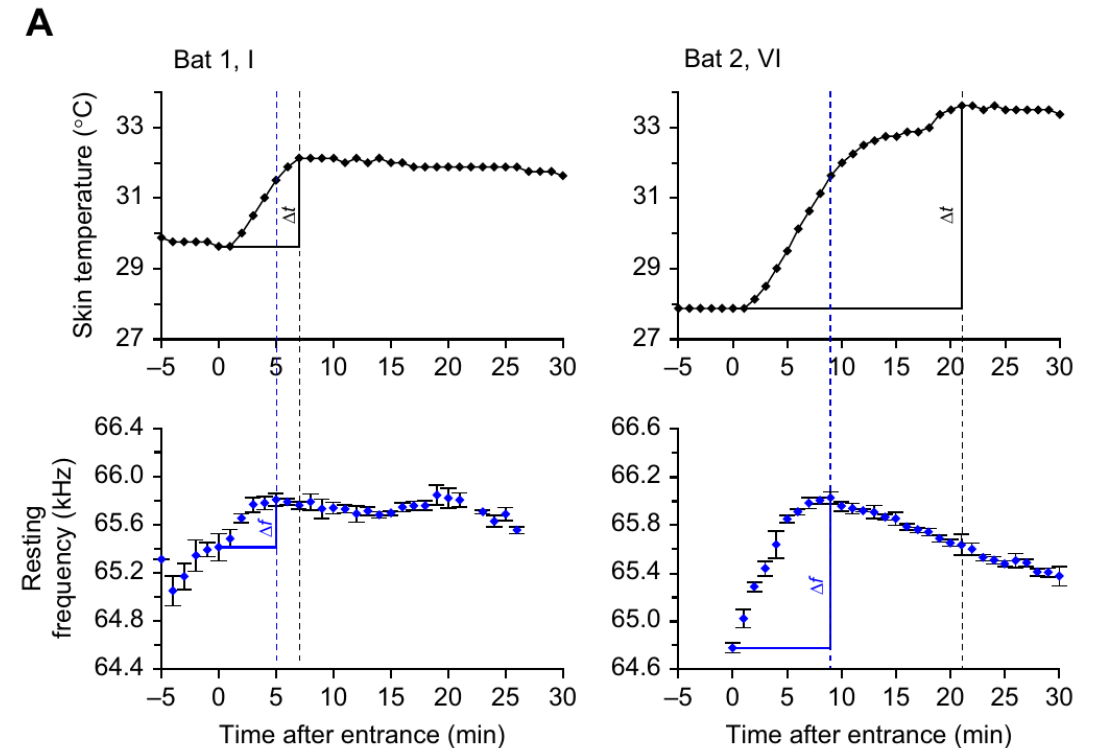
Schoeppler et al. (2022), Fig. 2(C–E). Skin temperature, calling activity, and resting CF2 frequency during activation.

Frequency can peak before measured skin temperature

Schoeppler and colleagues compared the time of maximum resting frequency with the time of maximum skin temperature during activation. The frequency maximum generally occurred earlier, while skin temperature continued increasing.

After the skin-temperature maximum, both temperature and resting frequency tended to decline. The timing difference prevents a simple conversion of every skin-temperature change into a fixed frequency change throughout warming and cooling.

The authors discussed delayed warming of skin relative to internal tissues as one explanation. Cochlear temperature was not recorded, so that explanation remains indirect. A correlation between two time series does not establish that their maxima must coincide or identify the precise thermal mechanism controlling vocal frequency.



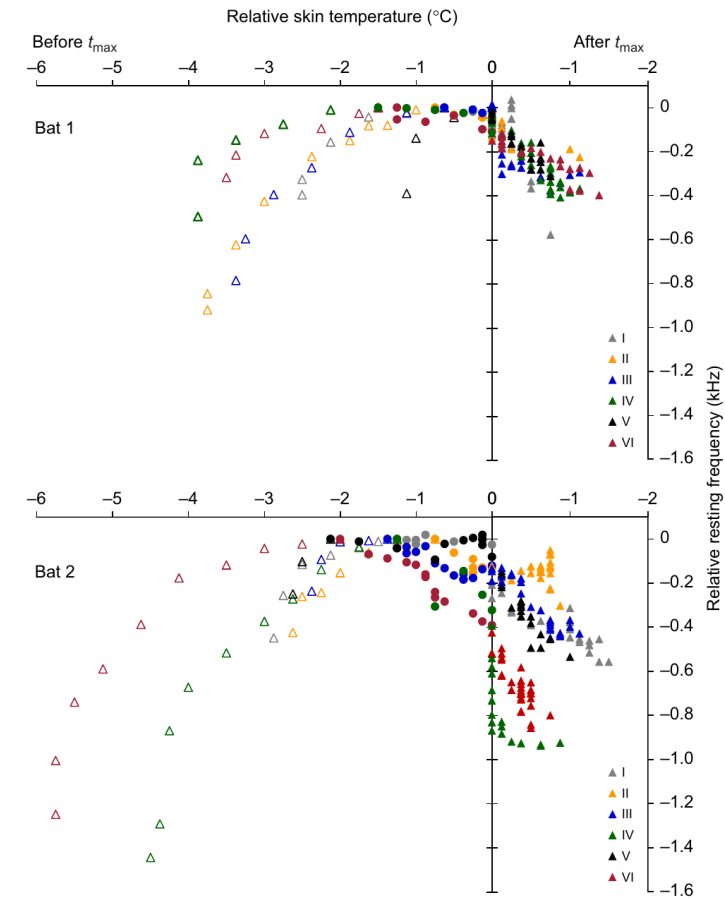
Schoeppler et al. (2022), Fig. 3(A). Resting-frequency and skin-temperature peaks differ in time during activation.

A changing sensory band can coexist with precise feedback

State-dependent resting-frequency changes and the coupling of resting and reference frequencies support an **adaptive operating band**: the feedback system can regulate around a preferred frequency that changes with the animal's physiological state.

Schoeppler and colleagues proposed that the controlled variable is activation of the cochlear foveal resonance system. **Afferent pathways** carry sensory activity toward control centers; **efferent pathways** carry motor commands toward the vocal apparatus, changing the next emitted call.

The warming experiment measured skin temperature and emitted frequency, not cochlear activation or a complete neural circuit. Its proposed control account must therefore be distinguished from the directly observed frequency shifts. The behavioral and peripheral studies together constrain how a moving sensory target can remain compatible with precise echo regulation.



Schoeppler et al. (2022), Fig. 5. Resting frequency covaries with skin temperature across activation and recovery.

Key takeaways

Echo frequency is the controlled signal. CF-FM bats lower emitted CF frequency as approach raises echo frequency, maintaining echoes near an individual reference band.

Audible feedback acts in both directions. Horseshoe bats lower calls for higher echoes and can raise calls for sufficiently audible lower echoes, with unequal correction ranges.

Neural codes include timing and population location. Midbrain neurons can retain narrow duration tuning near resonance; cortical populations organize frequency and preferred amplitude.

Resting and reference frequencies differ. The resting emission baseline usually lies slightly below the compensated echo reference; their offset must be measured at the same time.

The auditory fovea begins peripherally. Cochlear resonance and sharply tuned peripheral neurons concentrate sensitivity and selectivity near the CF2 echo band.

Precision can track a changing operating band. Resting and reference frequencies can shift together, and resting calls change during warming without a flight-induced Doppler shift.